

x800 Data Manager

Host Interface Manual

For IHE Laboratory Analytical Workflow (LAW) profile

Version 5.3

Software version 2.1 and higher



Publication information

Publication version	Software version	Revision date	Change description
1.0	1.4	October 2021	First version
2.0	1.5	January 2022	The SCoV2-QL and HIV-PSC assays are now supported.
3.0	1.5	April 2022	The CMV, EBV and BKV assays are now supported.
4.0	1.5	October 2022	The HBV-RNA, MAI, MTB, MTB-RIF/INH, and SARS-CoV-2 Duo assays are now supported. Sample type "Self, vaginal - RCCM/PC" has been added for assays HPV-GT and HPV-HR.
4.1	1.5	December 2022	Minor changes.
4.2	1.5	October 2023	What is new in publication version 4.2 (20)
4.3	2.0	April 2024	What is new in publication version 4.3 (19)
4.4	2.0	June 2024	What is new in publication version 4.4 (19)
4.5	2.0	August 2024	What is new in publication version 4.5 (18)
4.6	2.0	November 2024	What is new in publication version 4.6 (18)
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5.1	2.1	December 2024	What is new in publication version 5.1 (14)
5.2	2.1 and 2.2	August 2025	What is new in publication version 5.2 (13)
5.3	2.1 and higher	November 2025	What is new in publication version 5.3 (12)

Revision history

Edition notice

This publication is intended for Host Interface programmers and Roche field engineers of the x800 Data Manager.

Every effort has been made to ensure that all the information contained in this publication is correct at the time of publishing. However, the manufacturer of this product may need to update the publication information as output of product surveillance activities, leading to a new version of this publication.

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The **User Assistance** contains all information about the product, including the following:

- Routine operation
- Maintenance
- Safety
- Troubleshooting information
- Configuration information
- Background information
- Approvals
- Contact addresses

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The **User Guide** focuses on routine operation and maintenance. The chapters are organized according to the normal operation workflow.

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Disclaimer Some assays mentioned in this document may not be approved for distribution in all countries - consult your country's regulatory department for more information.

Updates to the Host Interface Manual may require updates to the LIS drivers.

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Preface

This document provides a basis for the development of communication between the x800 Data Manager and Laboratory Information Systems (LIS) using the HL7 communication protocols.

This document covers the generic structure of the communication between the x800 Data Manager and the LIS. Customer-supported Laboratory Information Systems are not part of the x800 Data Manager package.

This document describes only the segments and segment fields which are used in communication between the x800 Data Manager and the LIS. All other HL7 segments and segment fields are still valid but are not used by the x800 Data Manager.

Intended use

The **cobas**[®] 5800 System supports an automated and integrated workflow to run Polymerase Chain Reaction (PCR) based Nucleic Acid Testing (NAT) for use by trained professionals in laboratory settings.

The **cobas**[®] 5800 System combines the functionalities of instrumentation, consumables, reagents and data management to provide an efficient workflow from sample processing to result interpretation.

The **cobas**[®] 6800/8800 Systems support an automated and integrated workflow to run Polymerase Chain Reaction (PCR) based Nucleic Acid Testing (NAT).

The **cobas**[®] 6800/8800 Systems combine the functionalities of instrumentation, consumables, reagents, and data management to provide an efficient workflow from sample processing to result interpretation.

Symbols and abbreviations

Product names

Except where the context clearly indicates otherwise, the following product names and abbreviations are used.

Product name	Abbreviation
x800 Data Manager	software

☐ Product names

Symbols used in the publication

Symbol	Explanation
☎	Contact. Used in the User Assistance. Functionality currently unavailable.
🔗	Code example. Used in code titles and cross-references to codes.
•	List item
💡	Tip. Extra information on correct use or useful hints.
📖	Related topics containing further information
🖼	Figure. Used in figure titles and cross-references to figures.
☐	Table. Used in table titles and cross-references to tables.

☐ Symbols used in the publication

Abbreviations

The following abbreviations are used.

Abbreviation	Definition
ASAP	Assay Specific Analysis Package
ASCII	American Standard Code for Information Interchange
GUID	Global Unique Identifier
HL7	Health Level 7
IHE	Integrating the Healthcare Enterprise
ISO	International Standards Organization
IVD	In vitro diagnostic
LAW	Laboratory Analytical Workflow (supports integration of a host system in an IVD laboratory environment.)
LIS	Laboratory Information System
LOINC	Logical Observation Identifiers Names and Codes
MLLP	Minimal Lower Layer Protocol
OSI	Open System Interconnection

☐ Abbreviations

Abbreviation	Definition
QC	Quality Control
UCUM	Unified Code for Units of Measure
UTC	Coordinated Universal Time

☰ Abbreviations

What is new in publication version 5.3

Publication version 5.3 describes software versions 2.1 and higher.

New assays

The MPX-E and HDV assays are now supported.

-  MPX-E - LIS mapping (121)

-  HDV - LIS mapping (150)

Minor changes

A description of the test order submission message definition was added for software version 2.2 and higher.

-  Test order submission (OML^O33) (56)

The SPM-11-1 description for test order cancel was updated.

-  SPM - sample segment (70)

The OBX-5 optional value for test result was updated.

-  OBX - observation result segment for target result (93)

The notes about the Universal Service Identifier were updated.

-  RESP-4FLEX - LIS mapping (162)

-  RESP-FLEX - LIS mapping (164)

A new sample type Cell Collection Medium PLUS was added for HPV-GT and HPV-HR.

-  HPV-GT - Sample types and input volume (177)

-  HPV-HR - Sample types and input volume (179)

Several error flags were added (Q07D, U08, U12M, P07T, P09T, P19A, Y02T, Y03, Y04P, Y06A, Y07, Y46, Y49P).

Descriptions for several error flags were updated.

The description for C01X has been split to C01 and C01[H1...H2] for better understanding. No change to the software.

The description for C02X has been split to C02 and C02[H1...H2] for better understanding. No change to the software

-  Error flags (199)

What is new in publication version 5.2

	Publication version 5.2 describes software versions 2.1 and 2.2.
New assays	The BV/CV and RESP-4FLEX assays are now supported. • RESP-4FLEX - LIS mapping (162) • BV/CV mapping (173)
Message examples	Messages examples were removed. • Examples (193)
Minor changes	SPM-31-4 subcomponents were removed from all workflows. Segment descriptions and examples were updated. • HL7 message definition (43)

What is new in publication version 5.1

Publication version 5.1 describes software version 2.1.

New assays

The WNV, DPX, Babesia, ZIKA, and CHIKV/DENV assays are now supported.

- [WNV - LIS mapping \(125\)](#)
- [DPX - LIS mapping \(127\)](#)
- [Babesia - LIS mapping \(129\)](#)
- [ZIKA - LIS mapping \(131\)](#)
- [CHIKV/DENV - LIS mapping \(133\)](#)

New sample types

Additional sample types and volumes were added for UCAP.

- [Sample types and input volume \(190\)](#)

New message examples

Messages examples were added.

- [Examples \(193\)](#)

Minor changes

A reference in the Test order delete notification table was removed.

- [Supported workflows \(28\)](#)

Added DTM to HL7 data types table.

- [About HL7 elements \(37\)](#)

Examples were added to MSH-9 fields.

Segment descriptions and examples were updated.

- [HL7 message definition \(43\)](#)

A cross-reference was added.

- [Cancel \(OUL^R22\) \(101\)](#)

Some assay names were updated.

Table headings were updated.

- [Assays \(111\)](#)

The volume for PLASMA in the MPX sample types and input volume table was updated.

- [Sample types and input volume \(123\)](#)

Universal Service Identifier for CHIKV/DENV was corrected.

- [CHIKV/DENV - LIS mapping \(133\)](#)

Error flag CO2 was corrected.

•  Error flags (199)

What is new in publication version 5.0

	Publication version 5.0 describes software version 2.1.
Intended use	The Intended Use statement was expanded for the cobas® 6800/8800 Systems. •  Intended use (9)
Response to a test order submission (ORL^O34)	A new footnote was added to the message definition stating that the response (ORL^O34) excludes the SAMPLE group (including the ORDER group) in case of an error (MSA-1= AE) or when no test order has been assigned to the sample in the test order (OML^O33). •  Message definition (75)
Test result (OUL^R22)	A TCD test code segment was added after the first OBX segment of test result message and the INV segment was moved. The TCD segment includes the sample volume (Processing Input Volume) information. •  Message definition (86) •  TCD - test code details segment (95)
Blood screening assays	A note was added that blood screening assays are designed for Individual Donor Testing (IDT) and that they do not support pooling workflows. •  Blood screening (116)
Message examples	The message examples in the Examples chapter have been removed from this manual because they are not yet available for the x800 Data Manager software version 2.1. New message examples will be added in the upcoming version of this manual.
Minor changes	A Respiratory flex specific query parameter (WOS_ADDITIONAL) was added to the QPD-1.1 component description. •  QPD - query parameter segment (49)

Changed some brackets in the following messages:

- Message definition (51)
- Message definition (56)
- Message definition (63)
- Message definition (68)
- Message definition (75)
- Message definition (82)
- Message definition (86)
- Message definition (103)

Usage and/or cardinality changed for the following segments:

- ERR segment of the Request acknowledge (RSP^K11) message.
- NTE segment of the Test order submission (OML^O33) message.
- TCD segment of the Test order cancel (OML^O33) message.
- SAC segment of the Response to a test order submission (ORL^O34) message.
- ERR segment of the Result Accepted (ACK^R22) message.
- Message definition (51)
- Message definition (56)
- Message definition (68)
- Message definition (75)
- Message definition (103)

Statement changed for the RESP-FLEX assay.
This assay does not have a common Universal Service Identifier. For each test ORDER group, the software returns a result ORDER group in the OUL^R22 message. This is different compared to other multi channel assays (e.g. MPX) where the RESULT group is returned multiple times.

The footnote about the known defect was removed from the RESP-FLEX control results table.

- RESP-FLEX - LIS mapping (164)

What is new in publication version 4.6

Publication version 4.6 describes software version 2.0.

New assays

The WNV, DPX, Babesia, ZIKA, and CHIKV/DENV assays are now supported.

- [WNV - LIS mapping \(125\)](#)
- [DPX - LIS mapping \(127\)](#)
- [ZIKA - LIS mapping \(131\)](#)
- [CHIKV/DENV - LIS mapping \(133\)](#)

Minor changes

Removed old What is new sections.

Corrected Universal Service Identifier for CHIKV/DENV.

- [CHIKV/DENV - LIS mapping \(133\)](#)

Corrected error flag CO2.

- [Error flags \(199\)](#)

What is new in publication version 4.5

Publication version 4.5 describes software version 2.0.

New assays

The HEV, MPX, and RESP-FLEX assays are now supported.

- [HEV - LIS mapping \(117\)](#)
- [MPX - LIS mapping \(123\)](#)
- [RESP-FLEX - LIS mapping \(164\)](#)
- [Examples \(193\)](#)

Response to a test order submission (ORL^O34)

Usage and cardinality changed inside the ORDER group of the ORL message.

- [Response to a test order submission \(ORL^O34\) \(75\)](#)

Assay definitions are case-sensitive

Tip was added about the case-sensitivity of the assay definitions (test and target names and sample type names).

- [Introduction \(115\)](#)

What is new in publication version 4.4

	Publication version 4.4 describes software version 2.0.
New assay	The Malaria assay is now supported. • Malaria - LIS mapping (119) • Examples (193)
Minor changes	Additional references were added to the tables in the Supported workflows section. • Supported workflows (28) Descriptions of SPM-4 were updated in Test order submission (OML^O33), Test order cancel (OML^O33), Response to a test order submission (ORL^O34), and Test result (OUL^R22). Reference columns were added to all Message definition tables. • HL7 message definition (43) An HL7 message for Response to an unavailable test order was added. • Response to an unavailable test order (ORL^O34) (82) The appendix was restructured and assays were categorized in 4 sub-sections. • Appendix (109)

What is new in publication version 4.3

	Publication version 4.3 describes software version 2.0.
New assays	The HDV (RUO) and HCV-PSC assays are now supported. • HCV-PSC - LIS mapping (148) • HDV-RUO - LIS mapping (152)
Additional sample types	Sample types Diluted in cobas® MIS, PSC, RAW SPUTUM, SEDIMENT, and WHOLE BLOOD have been added.

Minor changes

Segment descriptions were updated in the HL7 message definition section.

- HL7 message definition (43)

Values in the Sample result codes for OBX segment and Control results tables were updated.

- Infectious disease (135)

The Workflow Prerequisites introduction was updated.

- Prerequisites (27)

The TCP keepalive section was updated.

- TCP keepalive (36)

What is new in publication version 4.2

Publication version 4.2 describes software version 1.5.

New assays

The SCoV2-FluA/B / SARS-CoV-2 & Influenza A/B v2 assays are now supported.

- SCoV2-FluA/B - LIS mapping (168)

Additional sample type

Sample type "Self, vaginal (for US only)" has been added for assays HPV-GT and HPV-HR.

- HPV-GT - LIS mapping (177)

- HPV-HR - LIS mapping (179)

Minor changes

A new section, Roche-provided hardware firewall was added to the Publication information.

New text was added to the Disclaimer section.

LAW and LDA profile sections were removed from the Publication information.

Abbreviations were updated.

Segment descriptions were updated in the HL7 message definition section.

Sample result codes for OBX segment and Tests and target tables were updated.

Errors in HPV-GT and HPV-HR Control results were corrected.

A typo in assay name MTB-RIF was corrected.

Message examples were updated.

-  Publication information (2)
-  Symbols and abbreviations (10)
-  HL7 message definition (43)
-  Infectious disease (135)
-  HPV-GT - LIS mapping (177)
-  HPV-HR - LIS mapping (179)
-  MTB-RIF/INH - LIS mapping (185)
-  Examples (193)

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Workflows

This chapter describes the workflow of communication between the x800 Data Manager and the LIS.

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Prerequisites

Communication protocol

The connection between the Analyzer and the Analyzer Manager is only established by the Data Manager (Analyzer). The Analyzer Manager (LIS) acts as the server.

The software uses the HL7 protocol version 2.5.1 to communicate with the LIS. The ASTM protocol is not supported.

 All HL7 communication between the software and the LIS must be ASCII code plain text.

Connection set-up

The software connects to the LIS through a Roche provided firewall. A Roche Service representative configures the firewall according to local customer network constraints.

The software is by default configured to connect to a LIS server on Port 3120. The software does not accept any incoming connections.

 Firewall security policy authorizes communication only via pre-established IP addresses.

For more information about firewall configuration, refer to the *Software Installation Manual*.

Supported workflows

The software supports 3 basic workflows to communicate with the LIS.
The IHE Laboratory Analytical Workflow (LAW) profile is followed.

Solicited test order submission

The software requests a test order from the LIS for each sample after sample recognition. The LIS sends the requested test order.

Direction	HL7 message	LAB workflow	Reference
Software→LIS	QBP^Q11	LAB-27 Test order query	✎ Work order request (QBP^Q11) (47)
Software←LIS	RSP^K11		✎ Request acknowledge (RSP^K11) (51)
Software←LIS	OML^O33	LAB-28 Test order	✎ Test order submission (OML^O33) (56)
			✎ Test order submission - no test order available (OML^O33) (63)
Software→LIS	ORL^O34		✎ Response to a test order submission (ORL^O34) (75)
			✎ Response to an unavailable test order (ORL^O34) (82)
☒ Test order query			

Result upload

The software sends test results to the LIS.

Direction	HL7 message	LAB workflow	Reference
Software→LIS	OUL^R22	LAB-29 Result	📄 Test result (OUL^R22) (86)
Software←LIS	ACK^R22		📄 Result Accepted (ACK^R22) (103)
📄 Result upload			

Test order delete notification

The software notifies the LIS when a test order is deleted.

Direction	HL7 message	LAB workflow	Reference
Software→LIS	OUL^R22	LAB-29 Test order deletion	✎ Cancel (OUL^R22) (101)
Software←LIS	ACK^R22		✎ Result Accepted (ACK^R22) (103)
☒ Test order delete notification			

Known restrictions

Character limitations

In HL7 messages, the characters | ^ ~ \ & are used (in this order) as delimiters and are not configurable.

Any character used as a delimiter should not be used in sample barcodes, or any manual text entry field (e.g. comments).

HL7 reference

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HL7 transport protocol

This chapter presents the lower layers of the HL7 protocol, as used by the x800 Data Manager.

In this chapter

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Physical communication

The standard HL7 refers to the highest level of the Open System Interconnection (OSI) model of the International Standards Organization (ISO). The HL7 Standard is primarily focused on the issues that occur within the seventh, or application, level. These are the definitions of the data to be exchanged, the timing of the exchanges, and the communication of certain application-specific errors between the applications. This chapter gives some recommendations for how to use HL7 with x800 Data Manager.

Minimal layer protocol

The Minimal Lower Layer Protocol (MLLP) is the most common standard for transmitting HL7 messages via TCP/IP over a local network.

A wrapping protocol is required to identify the start and the end of an HL7 message because TCP/IP is a continuous stream of bytes.

When using MLLP, an HL7 message is sent in one block, from the <Start block> <VT> until the <End block> including the final <CR>. A header and a trailer wrap the message block to mark its beginning and end. Headers and trailers are not part of the message's content.

```
<Start block>Data<End block><CR>
```

<Start block>	Start block character (1 byte) ASCII <VT>, i.e. <0x0B>. Do not confuse with the ASCII characters <SOH> or <STX>.
Data	Data (variable number of bytes) can contain any displayable ASCII characters and the carriage return character <CR>.
<End block>	End block character (1 byte) ASCII <FS>, i.e. <0x1C>. Do not confuse with the ASCII characters <ETX> or <EOT>.
<CR>	Carriage Return (1 byte) The ASCII carriage return character, i.e. <0x0D>.

Sending mode

The software sends one message at a time via a single connection.

- *Sequential sending:* After sending a message, the software waits for the acknowledgment from the host. If the host does not reply within a configurable timeout (30 to 90 seconds), the software marks the message as "*transmission failed*" and sends the next message.
- *Parallel sending:* By default, the software sends messages one after the other without waiting for the individual acknowledgment as long as 5 or less acknowledgments are pending. If 5 acknowledgments are pending, the software must wait for an acknowledgment before sending the next message.

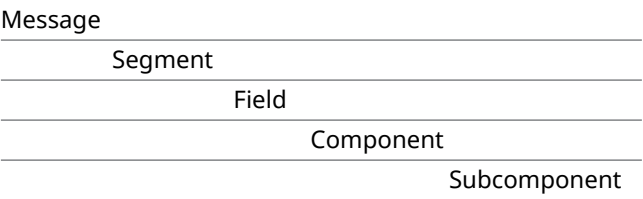
TCP keepalive

The keepalive functionality is active by default on the x800 Data Manager. After an idle time of 285 seconds it starts sending an empty TCP message every 30 seconds to the LIS.

No specific implementation on the LIS application level is required to handle the TCP keepalive message as TCP stack implementations already include the functionality.

About HL7 elements

Data in an HL7 message is organized hierarchically.



HL7 hierarchy

Message	HL7 defines different types of messages for specific purposes. A message is defined as an entire unit of data transmitted between sender and receiver. HL7 messages are written by a non-XML based language. A message is triggered by a real event which generates a workflow (e.g. availability of a test result to be sent to the host).
Segment	Each line of an HL7 message is a segment. Segments can be required or optional, and occur only once or repeatedly in a message. Each segment starts with a 3-character string (ID) that defines the segment type. The ID contains only printable characters (ASCII 32–126).
Field	Each segment consists of fields. A field contains information about the purpose of the segment. Fields may be required, optional, or conditional, and may occur only once or repeatedly.
Component and subcomponent	Fields are subdivided into components that refer to a logical set of characters within a field. Components can be further divided into subcomponents.
Delimiters	Encoding characters, or delimiters, define how data is separated in an HL7 message. The following delimiters are supported as specified in the MSH segment.

Delimiter	Function
	Separates fields
^	Separates components
&	Separates subcomponents
~	Separates repetitions
\	Indicates the escape character

HL7 delimiters

Data type

HL7 data types define the type of data that can be included in a field. A data type may contain additional data types that are referenced as components or subcomponents. 2–3 letter codes are used to specify the data type. Supported data types by the software are listed below.

Data type	Description
CE	Coded Element
DTM	Date/Time. x800 Data Manager sends the DTM object in UTC and the offset needs to be applied to convert to local time. String in the format YYYYMMDDhhmmss+-hhmm All other timestamps in the message shall relate to the time zone specified here. • YYYY = Year (four digits) • MM = Month (two digits) • DD = Day (two digits) • hh = Hour in UTC (two digits) • mm = Minute in UTC (two digits) • ss = Second in UTC (two digits) • +-hhmm = Time zone offset indicates whether the local time is ahead (+) or behind (-) UTC. E.g. 19981004010159+0100 for 2:01:59 on October 4, 1998 in Amsterdam, NL. (Time zone=+0100).
EI	Entity Identifier
EIP	Entity Identifier Pair
ID	Coded Values for HL7 tables
NA	Numeric Array
NM	A numeric value. Positive (+) or negative (-) is indicated before the numeric value. If it is not indicated, it is treated as positive (+). If a decimal point is not included, the numeric value is treated as an integer. There are no restrictions for placing '0' in the front or for multiple '0' placed at the end of numbers with decimal points.
ST	An alphanumeric string with a maximum length of 199 if not otherwise stated.
TS	Time Stamp. The default format is YYYYMMDDHHMMSS • YYYY is the four-digit Gregorian year • MM is the month • DD is the day. • HH is the hour, as 24-hour military time. • MM is the minute. • SS is the second. For example, 11:53 a.m. and 3 seconds, on September 5, 2015 is indicated 20150905115303.
TX	Text Data
XCN	Extended Composite ID Number and name for persons

 HL7 data types

**Transmitting the null value**

The "null" value is transmitted as 2 double quote marks (""), which is relevant when a message is sent to update a record. If no value is sent (i.e. omission of an optional field), the old value remains unchanged. If the null value is sent, the old value is changed to null (zero).

Color-coding

This publication visualizes the directionality of HL7 messages or message elements by different colors as illustrated below:

Incoming message (software<-host)

Outgoing message (software->host)

Message with unspecified directionality

HL7 cardinality and usage

Both cardinality and usage determine the appearance of HL7 elements (segment, field, component) in a message.

Cardinality

Cardinality identifies the minimum and maximum number of repetitions for a particular HL7 element (segment group, segment, or field). Cardinalities are expressed as a minimum-maximum pair of non-negative integers.

Value	Description
[0..0]	Element is never present.
[0..1]	Element may be omitted and can have at most 1 occurrence.
[1..1]	Element must have exactly 1 occurrence .
[0..n]	Element may be omitted or may repeat up to n times.
[1..n]	Element must appear at least once and may repeat up to n times.
[0..*]	Element may be omitted or repeat for a non-specified number of times ,
[1..*]	Element must appear at least once, and may repeat for a non-specified number of times.

 HL7 cardinality

Usage

Usage further defines the circumstances under which an element appears in a message. Some elements must always be present, others may never be present, and others may only be present under certain circumstances. The following codes are used to define the usage of an element.

Value	Description
R	Required
RE	Required but may be empty
O	Optional
C	Conditional
CE	Conditional but may be empty
-	Not supported

 HL7 usage

In addition, square and curly brackets are used to indicate the appearance and repetition of segments and segment blocks in the HL7 message structure trees.

Value	Description
[..]	0 or 1 occurrence
{..}	1 or multiple occurrences
[{..}]	0 or multiple occurrences

 Fields that are not supported by the software, are omitted in the following segment tables.

HL7 message definition

This chapter describes the structure of the supported HL7 messages by the software.

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Work order request (QBP^Q11)

Message definition

Query from x800 Data Manager ➔ LIS.

Segment	Description	Usage	Cardinality	Reference
MSH	Message header	R	[1..1]	MSH - message header segment (47)
QPD	Query parameter definition	R	[1..1]	QPD - query parameter segment (49)
RCP	Response Control Parameter	R	[1..1]	RCP - response control parameter segment (50)

QBP^Q11 message

MSH - message header segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-1		Field Delimiter	ST	1	1:1	R	The character specified in MSH-1 defines the field separator to be used for the rest of the message. The value is: (ASCII 124).
MSH-2		Encoding Characters	ST	4	1:1	R	The field contains the 4 delimiters in the following order: component separator, repetition separator, escape character, and subcomponent separator. The required values are: ^~\& (i.e. ASCII 94, 126, 92, and 38).
MSH-3		Sending Application	HD	227	1:1	R	The field uniquely identifies the sending application. The LIS server defines its content.
	MSH-3-1	Namespace ID	IS	20	1:1		Configurable in Host Settings , by default X800DM.
MSH-4		Sending Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-4. The software does not process the content of this field.
	MSH-4-1	Namespace ID	IS	20	1:1		Not used
MSH-5		Receiving Application	HD	227	1:1	R	The field uniquely identifies the receiving application. The LIS server defines its content.
	MSH-5-1	Namespace ID	IS	20	1:1		Configurable in Host Settings , by default Host.
MSH-6		Receiving Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-6. The software does not process the content of this field.
	MSH-6-1	Namespace ID	IS	20	1:1		Not used

MSH - message header segment (QBP^Q11)

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-7		Message Timestamp	TS	26	1:1	R	Date/time when the sending system created the message.
	MSH-7-1	Date/Time of Message Creation	DTM	19	1:1		String in the format YYYYMMDDhhmmss+-hhmm In this software version, DTM is represented in UTC with an (+/-) offset, not in local time. All other timestamps in the message shall relate to the time zone specified here. • YYYY = Year (four digits) • MM = Month (two digits) • DD = Day (two digits) • hh = Hour in UTC (two digits) • mm = Minute in UTC (two digits) • ss = Second in UTC (two digits) • +-hhmm = Time zone offset indicates whether the local time is ahead (+) or behind (-) UTC (e.g. 20120427123100+0100, indicates that UTC + 1 hour is the equivalent to the local time).
MSH-9		Message Type	MSG	15	1:1	R	Defines the type of message and trigger event Fixed value =QBP^Q11^QBP_Q11
	MSH-9-1	Message Code	IS	3	1:1		Fixed value =QBP
	MSH-9-2	Trigger Event	IS	3	1:1		Fixed value =Q11
	MSH-9-3	Message Structure	IS	7	1:1		Fixed value =QBP_Q11
MSH-10		Message Control ID	ST	36	1:1	R	The field contains a number or identifier that uniquely identifies the message. The receiving system returns this ID to the sending system in the MSA-2 segment (RSP^K11). e.g. GUID
MSH-11		Processing Mode	PT	3	1:1	R	
	MSH-11-1	Processing ID	ID	1	1:1		Fixed value = P (production)
MSH-12		HL7 Version ID	VID	60	1:1	R	The field specifies the version of the HL7 standard used in constructing, interpreting, and validating the messages.
	MSH-12-1	HL7 Version ID	ID	5	1:1		Fixed value =2.5.1
MSH-15		Accept Acknowledgment Type	ID	2	1:1	R	Conditions under which accept acknowledgments are required to be returned in response to this message. Fixed value =NE (never)
MSH-16		Application Acknowledgment Type	ID	2	1:1	R	Conditions under which application acknowledgments are required to be returned in response to this message. Fixed value =AL (always)
MSH-18		Character Set	ID	16	1:1	R	The field defines the character set used in the entire message. Fixed value =UNICODE UTF-8
MSH-21		Message Profile Identifier	EI	427	1:*	R	
	MSH-21-1	Entity ID	ST	50	1:1		Fixed value =LAB-27R
	MSH-21-2	Namespace ID	IS	20	1:1		Fixed value =ROCHE

MSH - message header segment (QBP^Q11)

QPD - query parameter segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
QPD-1		Query Message Name	CE	250	1:1	R	The field contains the name of the query. e.g. WOS_ROCHE^Work Order Step Roche Extension^ROCHE
	QPD-1.1	Identifier	ST	20	1:1		Supported values: - WOS_ADDITIONAL for reflex query - WOS_ROCHE for all other cases
	QPD-1.2	Text	ST	199	1:1		Fixed value = Work Order Step Roche Extension
	QPD-1.3	Coding System	ID	12	1:1		Fixed value = ROCHE
QPD-2		Query Tag	ST	32	1:1	C	e.g. GUID. When receiving the acknowledgment from the host (RSP^K11), this ID should be reflected in fields QAK-1 and QPD-2.
QPD-3		Container ID	EI	80	1:1	C	QPD-3 is used for sample identification and is matched against SAC-3 (Sample ID) in OML^O33.
	QPD-3-1	Entity Identifier	ST	50	1:1		Barcode ID (equal to Sample ID).
QPD-4		Carrier ID	EI	80	1:1	C	Will be matched against SAC-10 from the subsequent test order (OML^O33).
	QPD-4-1	Entity Identifier	ST	23	1:1		ID of the loaded rack where the sample is placed.
QPD-5		Rack Position	NA	80	1:1	C	Will be matched against SAC-11 from the subsequent test order (OML^O33).
							 This field is a Roche-specific field.
	QPD-5-1	Position in Carrier	NM	16	1:1		Position of the sample in the carrier. Possible values: - c5800 Standard Carrier: 1 to 16 - Collection Media Carrier: 1 to 7 - RD5 racks: 1 to 5

QPD - query parameter segment

RCP - response control parameter segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
RCP-1		Query Priority	ID	1	1:1	R	Query Priority. Fixed = I (immediate)
RCP-3		Response Modality	CE	250	1:1	R	Response modality. e.g. R^Real Time^HL70394
	RCP-3-1	Identifier	ST	20	1:1		Fixed = R (real time)
	RCP-3-2	Text	ST	199	0:1		Fixed = Real Time
	RCP-3-3	Coding System	ID	12	1:1		Fixed = HL70394

📖 RCP - response control parameter segment

Example message

```
MSH|^~\&|X800 DM||HOST||20210915103321+0200||QBP^Q11^QBP_Q11|e6c753d2-c752-4de4-b86d-0a9507000ce0|P|
2.5.1|||NE|AL||UNICODE UTF-8||LAB-27R^ROCHE
QPD|WOS_ROCHE^Work Order Step Roche Extension^ROCHE|75566b64-33be-4a3d-a154-b895305b271c|sample123|S
54|14
RCP|I||R^Real Time^HL70394
```

🔗 Example QBP^Q11 message Test Order query

📖 Related topics

- [Workflow](#)
 - [Solicited test order submission \(28\)](#)

Request acknowledge (RSP^K11)

Message definition

Acknowledgment from LIS ↘ x800 Data Manager.

Segment	Description	Usage	Cardinality	Reference
MSH	Message Header	R	[1..1]	✎ MSH - message header segment (51)
MSA	Message Acknowledgment	R	[1..1]	✎ MSA - message acknowledge segment (53)
[{ERR}]	Message Error	C	[0..*]	✎ ERR - error segment (54)
QAK	Query Acknowledgment	R	[1..1]	✎ QAK - query acknowledge segment (53)
QPD	Query Parameter Definition*	R	[1..1]	✎ QPD - query parameter segment (54)

✎ RSP^K11 message

* The QPD segment is expected according to HL7 standard, but the software ignores the QPD segment and its content in an incoming RSP^K11 message.

MSH - message header segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-1		Field Delimiter	ST	1	1:1	R	The character specified in MSH-1 defines the field separator to be used for the rest of the message. The value is: (ASCII 124).
MSH-2		Encoding Characters	ST	4	1:1	R	The field contains the 4 delimiters in the following order: component separator, repetition separator, escape character, and subcomponent separator. The required values are: ^~\& (i.e. ASCII 94, 126, 92, and 38).
MSH-3		Sending Application	HD	227	1:1	R	The field uniquely identifies the sending application. The LIS server defines its content.
	MSH-3-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-5 of QBP^Q11
MSH-4		Sending Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-4. The software does not process the content of this field.
	MSH-4-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-6 of QBP^Q11
MSH-5		Receiving Application	HD	227	1:1	R	The field uniquely identifies the receiving application. The LIS server defines its content.
	MSH-5-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-3 of QBP^Q11
MSH-6		Receiving Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-6. The software does not process the content of this field.
	MSH-6-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-4 of QBP^Q11

✎ MSH - message header segment (RSP^K11)

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-7		Message Timestamp	TS	26	1:1	R	Date/time when the sending system created the message.
	MSH-7-1	Date/Time of Message Creation	DTM	19	1:1		String in the format <code>YYYYMMDDhhmmss+-hhmm</code> In this software version, DTM is represented in UTC with an (+/-) offset, not in local time. All other timestamps in the message shall relate to the time zone specified here. • <code>YYYY</code> = Year (four digits) • <code>MM</code> = Month (two digits) • <code>DD</code> = Day (two digits) • <code>hh</code> = Hour in UTC (two digits) • <code>mm</code> = Minute in UTC (two digits) • <code>ss</code> = Second in UTC (two digits) • <code>+-hhmm</code> = Time zone offset indicates whether the local time is ahead (+) or behind (-) UTC (e.g. <code>20120427123100+0100</code>, indicates that UTC + 1 hour is the equivalent to the local time).
MSH-9		Message Type	MSG	15	1:1	R	Defines the type of message and trigger event Fixed value = <code>RSP^K11^RSP_K11</code>
	MSH-9-1	Message Code	IS	3	1:1		Fixed value = <code>RSP</code>
	MSH-9-2	Trigger Event	IS	3	1:1		Fixed value = <code>K11</code>
	MSH-9-3	Message Structure	IS	7	1:1		Fixed value = <code>RSP_K11</code>
MSH-10		Message Control ID	ST	36	1:1	R	A number or identifier that uniquely identifies the message e.g. <code>GUID</code>
MSH-11		Processing Mode	PT	3	1:1	R	
	MSH-11-1	Processing ID	ID	1	1:1		Fixed value = <code>P</code> (production)
MSH-12		HL7 Version ID	VID	60	1:1	R	The version of the HL7 standard used in constructing, interpreting, and validating the messages.
	MSH-12-1	HL7 Version ID	ID	5	1:1		Fixed value = <code>2.5.1</code>
MSH-18		Character Set	ID	16	1:1	R	The character set used in the entire message. Fixed value = <code>UNICODE UTF-8</code>
MSH-21		Message Profile Identifier	EI	427	1:*	R	
	MSH-21-1	Entity ID	ST	50	1:1		Fixed value = <code>LAB-27R</code>
	MSH-21-2	Namespace ID	IS	20	1:1		Fixed value = <code>ROCHE</code>

MSH - message header segment (RSP^K11)

MSA - message acknowledge segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSA-1		Acknowledgment Code	ID	2	1:1	R	The following codes are supported: - AA = Application Accepted - AR = Application Rejected - AE = Application Error
MSA-2		Message Control ID	ST	50	1:1	R	Copy of MSH-10 of the request message (QBP^Q11).

MSA - message acknowledge segment

QAK - query acknowledge segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
QAK-1		Query Tag	ST	32	1:1	C	Copy of QPD-2 from the request message (QBP^Q11) e.g. GUID
QAK-2		Query Response Status	ID	2	1:1	R	Supported values: - OK = No errors - AE = Application Error - AR = Application Rejected
QAK-3		Message Query Name	CE	250	1:1	R	Copy of QPD-1 from the request message (QBP^Q11).
	QAK-3-1	Identifier	ST	20	1:1		Fixed value =WOS_ROCHE
	QAK-3-2	Text	ST	199	1:1		Fixed value =Work Order Step Roche Extension
	QAK-3-3	Coding System	ID	12	1:1		Fixed value =ROCHE

QAK - query acknowledge segment

QPD - query parameter segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
QPD-1		Query Message Name	CE	250	1:1	R	Copy of QPD-1 from the request message (QBP^Q11).
	QPD-1.1	Identifier	ST	20	1:1		Fixed value = WOS_ROCHE
	QPD-1.2	Text	ST	199	1:1		Fixed value = Work Order Step Roche Extension
	QPD-1.3	Coding System	ID	12	1:1		Fixed value = ROCHE
QPD-2		Query Tag	ST	32	1:1	C	Copy of QPD-2 from the request message (QBP^Q11) e.g. GUID
QPD-3		Container ID	EI	80	1:1	C	Barcode ID (equal to Sample ID)
	QPD-3-1	Entity Identifier	ST	50	1:1		Copy of QPD-3 from the request message (QBP^Q11).
QPD-4		Carrier ID	EI	80	1:1	C	Rack barcode.
	QPD-4-1	Entity Identifier	ST	23	1:1		Copy of QPD-4 from the request message (QBP^Q11).
QPD-5		Rack ID	NA	80	1:1	C	Position of sample within the rack
	QPD-5-1	Position in Carrier	NM	16	1:1		Copy of QPD-5 from the request message (QBP^Q11).

QPD - query parameter segment

ERR - error segment

Error segment if message acknowledge is negative (MSA-1 is unequal AA).

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
ERR-2		Error Location	ERL	18	0:*	O	
	ERR-2-1	Segment ID	ST	3	1:1		Example: QPD
	ERR-2-2	Segment Sequence	NM	2	1:1		Example: 1
	ERR-2-3	Field Number	NM	2	0:1		Example: 3
	ERR-2-4	Field Repetition	NM	2	0:1		Example: 1
	ERR-2-5	Component Number	NM	2	0:1		Example: 1
	ERR-2-6	Sub-Component Number	NM	2	0:1		Example: ""

ERR - error segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
ERR-3		Error Code	CWE	250	1:1	R	
	ERR-3-1	Identifier	ST	20	1:1		Value from the table HL70357. Limited to 100, 101, 102, 103, 200, 201, 202, 203, 207
	ERR-3-2	Text	ST	199	0:1		Not used
	ERR-3-3	Coding System	ID	12	1:1		Fixed value = HL70357
ERR-4		Severity	ID	1	1:1	R	Fixed value = E (error)
ERR-8		User Message	TX	250	0:1	O	Example: Barcode length less than 3 characters

ERR - error segment

Example messages

```
MSH|^~\&|HOST||X800 DM||20210915123329+0200||RSP^K11^RSP_K11|39a277e3-3b3a-43a7-a7ba-01b01e130216|P|
2.5.1|||||UNICODE UTF-8|||LAB-27R^ROCHE
MSA|AA|e6c753d2-c752-4de4-b86d-0a9507000ce0
QAK|75566b64-33be-4a3d-a154-b895305b271c|OK|WOS_ROCHE^Work Order Step Roche Extension^ROCHE
QPD|WOS_ROCHE^Work Order Step Roche Extension^ROCHE|75566b64-33be-4a3d-a154-b895305b271c|sample123|S
54|14
```

Example RSP^K11 message Acknowledge from host

Related topics

- Workflow
 - Solicited test order submission (28)

Test order submission (OML^O33)

Message definition

For software version 2.1, up to 3 independent assays per sample ID can be combined in a single test order message.

For software version 2.2 and higher, up to 6 independent assays per sample ID can be combined in a single test order message.

 An assay can have multiple tests, for example, RespFlex, CT/NG, DPX.

Order download from LIS  x800 Data Manager.

Segment	Description	Usage	Cardinality	Reference
MSH	Message header	R	[1..1]	✎ MSH - message header segment (57)
	--- SAMPLE start	R	[1..1]	
SPM	Sample information	R	[1..1]	✎ SPM - sample segment (58)
SAC	Container information	R	[1..1]	✎ SAC - sample container segment (59)
{	--- ORDER start	R	[1..*]	
ORC	Common order	R	[1..1]	✎ ORC - common order segment (60)
	--- OBSERVATION ORDER start	R	[1..1]	
OBR	Observation order	R	[1..1]	✎ OBR - observation request segment (60)
TCD	Test code detail	R	[1..1]	✎ TCD - test code details segment (61)
[{NTE}]	Notes and comments	O	[0..*]	✎ NTE - notes and comments segment (62)
	--- OBSERVATION ORDER end			
}	--- ORDER end			
	--- SAMPLE end			

📋 OML^O33 message

MSH - message header segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-1		Field Delimiter	ST	1	1:1	R	The character specified in MSH-1 defines the field separator to be used for the rest of the message. The value is: (ASCII 124).
MSH-2		Encoding Characters	ST	4	1:1	R	The field contains the 4 delimiters in the following order: component separator, repetition separator, escape character, and subcomponent separator. The required values are: ^~\& (i.e. ASCII 94, 126, 92, and 38).
MSH-3		Sending Application	HD	227	1:1	R	The field uniquely identifies the sending application. The LIS server defines its content.
	MSH-3-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-5 of QBP^Q11.
MSH-4		Sending Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-4. The software does not process the content of this field.
	MSH-4-1	Namespace ID	IS	4	1:1		Should be a copy of MSH-6 of QBP^Q11.
MSH-5		Receiving Application	HD	227	1:1	R	The field uniquely identifies the receiving application. The LIS server defines its content.
	MSH-5-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-3 of QBP^Q11.
MSH-6		Receiving Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-6. The software does not process the content of this field.
	MSH-6-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-4 of QBP^Q11.
MSH-7		Message Timestamp	TS	26	1:1	R	Date/time when the sending system created the message.
	MSH-7-1	Date/Time of Message Creation	DTM	19	1:1		String in the format YYYYMMDDhhmmss+-hhmm In this software version, DTM is represented in UTC with an (+/-) offset, not in local time. All other timestamps in the message shall relate to the time zone specified here. • YYYY = Year (four digits) • MM = Month (two digits) • DD = Day (two digits) • hh = Hour in UTC (two digits) • mm = Minute in UTC (two digits) • ss = Second in UTC (two digits) • +-hhmm = Time zone offset indicates whether the local time is ahead (+) or behind (-) UTC (e.g. 20120427123100+0100, indicates that UTC + 1 hour is the equivalent to the local time).
MSH-9		Message Type	MSG	15	1:1	R	Defines the type of message and trigger event Fixed value =OML^O33^OML_O33
	MSH-9-1	Message Code	IS	3	1:1		Fixed value =OML
	MSH-9-2	Trigger Event	IS	3	1:1		Fixed value =O33
	MSH-9-3	Message Structure	IS	7	1:1		Fixed value =OML_O33

MSH - message header segment (OML^O33)

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-10		Message Control ID	ST	36	1:1	R	The field contains a number or identifier that uniquely identifies the message. The receiving system returns this ID to the sending system in the MSA-2 segment (ORL^O34). e.g. GUID
MSH-11		Processing Mode	PT	3	1:1	R	
	MSH-11-1	Processing ID	ID	1	1:1		Fixed value = P (production)
MSH-12		HL7 Version ID	VID	60	1:1	R	The field specifies the version of the HL7 standard used in constructing, interpreting, and validating the messages.
	MSH-12-1	HL7 Version ID	ID	5	1:1		Fixed value = 2.5.1
MSH-15		Accept Acknowledgment Type	ID	2	1:1	R	Conditions under which accept acknowledgments are required to be returned in response to this message. Fixed value = NE (never)
MSH-16		Application Acknowledgment Type	ID	2	1:1	R	Conditions under which application acknowledgments are required to be returned in response to this message. Fixed value = AL (always)
MSH-18		Character Set	ID	16	1:1	R	The field defines the character set used in the entire message. Fixed value = UNICODE UTF-8
MSH-21		Message Profile Identifier	EI	427	1:*	R	
	MSH-21-1	Entity ID	ST	50	1:1		Fixed value = LAB-28
	MSH-21-2	Namespace ID	IS	20	1:1		Fixed value = IHE

MSH - message header segment (OML^O33)

SPM - sample segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SPM-1		Set ID	SI	4	1:1	R	Fixed value = 1
SPM-2		Sample ID	EIP	80	1:1	R	Sample ID / tube barcode
	SPM-2-1	Placer Assigned Id		80	1:1		
	SPM-2-1-1	Entity ID	ST	50	1:1		Sample Barcode ID (e.g. 12345&ROCHE)
	SPM-2-1-2	Namespace ID	IS	20	0:1		Fixed value = ROCHE

SPM - sample segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SPM-4		Sample Type	CWE	250	1:1	R	The precise nature of the entity that is the source material for the observation (e.g. SER^serum^HL70487). See Sample types and input volume in Assays (111)
	SPM-4-1	Identifier	ST	20	1:1		Code for sample type (e.g. SER)
	SPM-4-2	Text	ST	199	0:1		Textual description of sample type (e.g. serum)
	SPM-4-3	Coding System	ID	12	1:1		99ROC or HL70487
SPM-11		Sample Role	CWE	250	1:1	R	The content of the field depends on the sample type defined in SPM-4. e.g. P^HL70369
	SPM-11-1	Identifier	ST	20	1:1	R	Supported values: • P = Patient sample • Q = Quality control sample
	SPM-11-2	Text	ST	199	0:1		
	SPM-11-3	Coding System	ID	12	1:1		Fixed value = HL70369
SPM-31		Other Sample ID	EIP	80	0:1	O	A secondary sample ID is additional information sent by the LIS which is returned in the result message (OUL) SPM-31. Example: 12345
	SPM-31-1	ID	ST	80	1:1	R	Sample barcode ID

SPM - sample segment

SAC - sample container segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SAC-3		Container ID	EI	80	1:1	R	Copy of QPD-3 of QBP^Q11
	SAC-3-1	Entity Identifier	ST	50	1:1		Specimen barcode
SAC-10		Carrier ID	EI	80	0:1	O	Copy of QPD-4 of QBP^Q11
	SAC-10-1	Entity Identifier	ST	20	1:1		Rack ID.
SAC-11		Position in Carrier	NA	80	0:1	O	Copy of QPD-5 of QBP^Q11
	SAC-11-1	Value 1	NM	16	1:1		Position in carrier.

SAC - specimen container segment

ORC - common order segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
ORC-1		Order Control	ID	2	1:1	R	Supported values: - NW = New order - CA = Order cancellation
ORC-9		Transaction Date/Time	TS	14	1:1	R	
	ORC-9-1	Timestamp	DTM	14	1:1		Date/Time at which the order was created. String in the format YYYYMMDDhhmmss (e.g. 20210915123330).

ORC - common order segment

OBR - observation request segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
OBR-2		Placer Order Number	EI	50	1:1	R	The LIS placer order number. The value is returned in the OBR-2 from the OUL message.
	OBR-2-1	Entity Identifier	ST	50	1:1		Unique host order id.
OBR-4		Universal Service Identifier	CE	250	1:1	R	Test (Assay) codes are specified in the ASAP. e.g. 42595-9^HBV^LN • See Test and targets in Assays (111)
	OBR-4-1	Identifier	ST	20	1:1		Test code (e.g. 42595-9).
	OBR-4-2	Text	ST	199	1:1		Test name (e.g. HBV).
	OBR-4-3	Coding System	ID	12	1:1		Example: LN for LOINC

OBR - observation request segment

TCD - test code details segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
TCD-1		Universal Service Identifier	CE	250	1:1	R	Test (Assay) codes are specified in the ASAP. e.g. 42595-9^HBV^LN See Test and targets in Assays (111)
	TCD-1-1	Identifier	ST	20	1:1		Test code (e.g. 42595-9).
	TCD-1-2	Text	ST	199	1:1		Test name (e.g. HBV or copy of OBR-4-1).
	TCD-1-3	Coding System	ID	12	1:1		Example: LN for LOINC
TCD-8		Analyte Repeat Status	CE	250	0:1	C	In case of a reflex order required (fixed value = F^Reflex test^HL70389), otherwise ignored.
	TCD-8-1	Identifier	ST	20	1:1		In case of reflex order, fixed value = F, otherwise empty.
	TCD-8-2	Text	ST	199	1:1		In case of reflex order, fixed value = Reflex test, otherwise empty.
	TCD-8-3	Coding System	ID	12	1:1		In case of reflex order, fixed value = HL70389, otherwise empty.
TCD-9		Specimen Consumption Quantity	CQ	242	1:1	R	Example: 850^uL&&UCUM
	TCD-9-1	Process Input Volume	NM	16	1:1		Example: 850
	TCD-9-2	Quantity Units	CE	225	1:1		
	TCD-9-2-1	Coded Unit	ST	20	1:1		Fixed value = uL
	TCD-9-2-2	Text	ST	199	1:1		Can be empty or a copy of TCD-9-2-1.
	TCD-9-2-3	Coding System	ID	4	1:1		Fixed value = UCUM

☐ TCD - test code details segment

NTE - notes and comments segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
NTE-1		Set ID	SI	4	1:1	R	Sequence Number. May be used where multiple NTE segments are included in a message. Their numbering must be described in the application message definition. The values range from 1 to 9999. Numbering must be restarted in each group.
NTE-2		Source of Comment	ID	1	1:1	R	Fixed value =A
NTE-3		Comment Text	FT	65536	1:1	R	e.g. Damaged barcode label replaced
NTE-4		Comment Type	CE	250	0:1	O	Ignored

📄 NTE - notes and comments segment

Example messages

```
MSH|^~\&|HOST||X800 DM||20210915123330+0200||OML^O33^OML_O33|0b070615-e2ec-4463-92db-3fd53b21daa3|P|2.5.1|||NE|AL||UNICODE UTF-8|||LAB-28^IHE  
SPM|1|sample123&ROCHE||UR^Urine^HL70487|||||P^HL70369  
SAC|||sample123|||||S54|14  
ORC|NW|||||20210915123330  
OBR||3821||72828-7^CT/NG^LN  
TCD|72828-7^CT/NG^LN|||||850^uL&&UCUM
```

📄 Example OML^O33 message Test Order submission

📄 **Related topics**

- [Workflows](#)
 - [Solicited test order submission \(28\)](#)

Test order submission - no test order available (OML^O33)

Message definition

Order download from LIS  x800 Data Manager

Segment	Description	Usage	Cardinality	Reference
MSH	Message header	R	[1..1]	✎ MSH - message header segment (63)
	--- SAMPLE start	R	[1..1]	
SPM	Sample information	R	[1..1]	✎ SPM - sample segment (65)
SAC	Container information	R	[1..1]	✎ SAC - specimen container segment (66)
	--- ORDER start	R	[1..1]	
ORC	Common order	R	[1..1]	✎ ORC - common order segment (66)
[{NTE}]	Notes and comments	O	[0..*]	✎ NTE - notes and comments segment (67)
	--- ORDER end			
	--- SAMPLE end			

✎ OML^O33 message

MSH - message header segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-1		Field Delimiter	ST	1	1:1	R	The character specified in MSH-1 defines the field separator to be used for the rest of the message. The value is: (ASCII 124).
MSH-2		Encoding Characters	ST	4	1:1	R	The field contains the 4 delimiters in the following order: component separator, repetition separator, escape character, and subcomponent separator. The required values are: ^~\& (i.e. ASCII 94, 126, 92, and 38).
MSH-3		Sending Application	HD	227	1:1	R	The field uniquely identifies the sending application. The LIS server defines its content.
	MSH-3-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-5 of QBP^Q11
MSH-4		Sending Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-4. The system does not process the content of this field.
	MSH-4-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-6 of QBP^Q11
MSH-5		Receiving Application	HD	227	1:1	R	The field uniquely identifies the receiving application. The LIS server defines its content.
	MSH-5-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-3 of QBP^Q11

✎ MSH - message header segment (OML^O33)

Roche Diagnostics

x800 Data Manager · Software version 2.1 and higher · Host Interface Manual · Version 5.3

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-6		Receiving Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-6. The system does not process the content of this field.
	MSH-6-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-4 of QBP^Q11
MSH-7		Message Timestamp	TS	26	1:1	R	Date/time when the sending system created the message.
	MSH-7-1	Date/Time of Message Creation	DTM	19	1:1		String in the format <code>YYYYMMDDhhmmss+-hhmm</code> In this software version, DTM is represented in UTC with an (+/-) offset, not in local time. All other timestamps in the message shall relate to the time zone specified here. • <code>YYYY</code> = Year (four digits) • <code>MM</code> = Month (two digits) • <code>DD</code> = Day (two digits) • <code>hh</code> = Hour in UTC (two digits) • <code>mm</code> = Minute in UTC (two digits) • <code>ss</code> = Second in UTC (two digits) • <code>+-hhmm</code> = Time zone offset indicates whether the local time is ahead (+) or behind (-) UTC (e.g. <code>20120427123100+0100</code>, indicates that UTC + 1 hour is the equivalent to the local time).
MSH-9		Message Type	MSG	15	1:1	R	Defines the type of message and trigger event Fixed value = <code>OML^O33^OML_O33</code>
	MSH-9-1	Message Code	IS	3	1:1		Fixed value = <code>OML</code>
	MSH-9-2	Trigger Event	IS	3	1:1		Fixed value = <code>O33</code>
	MSH-9-3	Message Structure	IS	7	1:1		Fixed value = <code>OML_O33</code>
MSH-10		Message Control ID	ST	36	1:1	R	The field contains a number or identifier that uniquely identifies the message. The receiving system returns this ID to the sending system in the MSA-2 segment (ORL^O34). e.g. <code>GUID</code>
MSH-11		Processing Mode	PT	3	1:1	R	
	MSH-11-1	Processing ID	ID	1	1:1		Fixed value = <code>P</code> (production)
MSH-12		HL7 Version ID	VID	60	1:1	R	The field specifies the version of the HL7 standard used in constructing, interpreting, and validating the messages.
	MSH-12-1	HL7 Version ID	ID	5	1:1		Fixed value = <code>2.5.1</code>
MSH-15		Accept Acknowledgment Type	ID	2	1:1	R	Conditions under which accept acknowledgments are required to be returned in response to this message. Fixed value = <code>NE</code> (never)
MSH-16		Application Acknowledgment Type	ID	2	1:1	R	Conditions under which application acknowledgments are required to be returned in response to this message. Fixed value = <code>AL</code> (always)
MSH-18		Character Set	ID	16	1:1	R	The field defines the character set used in the entire message. Fixed value = <code>UNICODE UTF-8</code>

MSH - message header segment (OML^O33)

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-21		Message Profile Identifier	EI	427	1:*	R	
	MSH-21-1	Entity ID	ST	50	1:1		Fixed value = LAB-28
	MSH-21-2	Namespace ID	IS	20	1:1		Fixed value = IHE

MSH - message header segment (OML^O33)

SPM - sample segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SPM-1		Set ID	SI	4	1:1	R	Fixed value = 1
SPM-4		Sample Type	CWE	250	1:1	R	
	SPM-4-1	Identifier	ST	20	1:1		Fixed value = NULL
SPM-11		Sample Role	CWE	250	1:1	R	The content of the field depends on the sample type defined in SPM-4.
	SPM-11-1	Identifier	ST	20	1:1		Fixed value = U (unknown)
	SPM-11-2	Text	ST	199	0:1		
	SPM-11-3	Coding System	ID	12	1:1		Fixed value = HL70369
SPM-31		Other Specimen ID	EIP	80	0:1	O	Can be used for a secondary sample ID (e.g. 12345).
	SPM-31-1	ID	ST	80	1:1	R	Specimen barcode ID.

SPM - sample segment

SAC - specimen container segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SAC-3		Container ID	EI	80	1:1	R	
	SAC-3-1	Entity Identifier	ST	50	1:1		Specimen barcode. Copy of QPD-3 of QBP^Q11.
SAC-10		Carrier ID	EI	80	0:1	O	Copy of QPD-4 of QBP^Q11.
	SAC-10-1	Entity Identifier	ST	20	1:1		Rack ID.
SAC-11		Position in Carrier	NA	80	0:1	O	Copy of QPD-5 of QBP^Q11.
	SAC-11-1	Value 1	NM	16	1:1		Position in carrier.

☐ SAC - specimen container segment

ORC - common order segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
ORC-1		Order Control	ID	2	1:1	R	Fixed value = DC (discontinue)
ORC-9		Transaction Date/Time	TS	14	1:1	R	
	ORC-9-1	Timestamp	DTM	14	1:1		Date/Time at which the order was created. String in the format YYYYMMDDhhmmss (e.g. 20180730123400)

☐ ORC - common order segment

NTE - notes and comments segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
NTE-1		Set ID	SI	4	1:1	R	Sequence Number. May be used where multiple NTE segments are included in a message. Their numbering must be described in the application message definition. The values range from 1 to 9999. Numbering must be restarted in each group.
NTE-2		Source of Comment	ID	1	1:1	R	Fixed value = A
NTE-3		Comment Text	FT	65536	1:1	R	e.g. No order for this barcode ID
NTE-4		Comment Type	CE	250	0:1	O	Ignored

■ NTE - notes and comments segment

Example message

```
MSH|^~\&|HOST||X800 DM||20210915123330+0200||OML^O33^OML_O33|0b070615-e2ec-4463-92db-3fd53b21daa3|P|
2.5.1|||NE|AL||UNICODE UTF-8|||LAB-28^IHE
SPM|1|||U^HL70369
SAC|||sample123|||S54|14
ORC|DC|||20210915123330
```

🔗 Example OML^O33 message Negative test order submission (no order)

Related topics

- [Workflows](#)
 - [Solicited test order submission \(28\)](#)

Test order cancel (OML^O33)

Message definition

Order download from LIS  x800 Data Manager

Segment	Description	Usage	Cardinality	Reference
MSH	Message header	R	[1..1]	✎ MSH - message header segment (68)
	--- SAMPLE start	R	[1..1]	
SPM	Sample information	R	[1..1]	✎ SPM - sample segment (70)
SAC	Container information	R	[1..1]	✎ SAC - specimen container segment (71)
	--- ORDER start	R	[1..1]	
ORC	Common order (for one battery)	R	[1..1]	✎ ORC - common order segment (71)
	--- OBSERVATION ORDER start	R	[1..1]	
OBR	Observation order	R	[1..1]	✎ OBR - observation request segment (72)
TCD	Test code detail	R	[1..1]	✎ TCD - test code details segment (72)
[{NTE}]	Notes and comments	O	[0..*]	✎ NTE - notes and comments segment (73)
	--- OBSEVATION ORDER end			
	--- ORDER end			
	--- SAMPLE end			

📋 OML^O33 message

MSH - message header segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-1		Field Delimiter	ST	1	1:1	R	The character specified in MSH-1 defines the field separator to be used for the rest of the message. The value is: (ASCII 124).
MSH-2		Encoding Characters	ST	4	1:1	R	The field contains the 4 delimiters in the following order: component separator, repetition separator, escape character, and subcomponent separator. The required values are: ^~\& (i.e. ASCII 94, 126, 92, and 38).
MSH-3		Sending Application	HD	227	1:1	R	The field uniquely identifies the sending application. The LIS server defines its content.
	MSH-3-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-5 of QBP^Q11.

📋 MSH - message header segment (OML^O33)

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-4		Sending Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-4. The system does not process the content of this field.
	MSH-4-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-6 of QBP^Q11.
MSH-5		Receiving Application	HD	227	1:1	R	The field uniquely identifies the receiving application. The LIS server defines its content.
	MSH-5-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-3 of QBP^Q11.
MSH-6		Receiving Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-6. The system does not process the content of this field.
	MSH-6-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-4 of QBP^Q11.
MSH-7		Message Timestamp	TS	26	1:1	R	Date/time when the sending system created the message.
	MSH-7-1	Date/Time of Message Creation	DTM	19	1:1		String in the format YYYYMMDDhhmmss+-hhmm In this software version, DTM is represented in UTC with an (+/-) offset, not in local time. All other timestamps in the message shall relate to the time zone specified here. • YYYY = Year (four digits) • MM = Month (two digits) • DD = Day (two digits) • hh = Hour in UTC (two digits) • mm = Minute in UTC (two digits) • ss = Second in UTC (two digits) • +-hhmm = Time zone offset indicates whether the local time is ahead (+) or behind (-) UTC (e.g. 20120427123100+0100, indicates that UTC + 1 hour is the equivalent to the local time).
MSH-9		Message Type	MSG	15	1:1	R	Defines the type of message and trigger event
	MSH-9-1	Message Code	IS	3	1:1		Fixed value = OML
	MSH-9-2	Trigger Event	IS	3	1:1		Fixed value = O33
	MSH-9-3	Message Structure	IS	7	1:1		Fixed value = OML_O33
MSH-10		Message Control ID	ST	36	1:1	R	The field contains a number or identifier that uniquely identifies the message. The receiving system returns this ID to the sending system in the MSA-2 segment (ORL^O34). e.g. GUID
MSH-11		Processing Mode	PT	3	1:1	R	
	MSH-11-1	Processing ID	ID	1	1:1		Fixed value = P (production)
MSH-12		HL7 Version ID	VID	60	1:1	R	The field specifies the version of the HL7 standard used in constructing, interpreting, and validating the messages.
	MSH-12-1	HL7 Version ID	ID	5	1:1		Fixed value = 2.5.1
MSH-15		Accept Acknowledgment Type	ID	2	1:1	R	Conditions under which accept acknowledgments are required to be returned in response to this message. Fixed value = NE (never)

MSH - message header segment (OML^O33)

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-16		Application Acknowledgment Type	ID	2	1:1	R	Conditions under which application acknowledgments are required to be returned in response to this message. Fixed value =AL (always)
MSH-18		Character Set	ID	16	1:1	R	The field defines the character set used in the entire message. Fixed value =UNICODE UTF-8
MSH-21		Message Profile Identifier	EI	427	1:*	R	
	MSH-21-1	Entity ID	ST	50	1:1		Fixed value =LAB-28
	MSH-21-2	Namespace ID	IS	20	1:1		Fixed value =IHE

MSH - message header segment (OML^O33)

SPM - sample segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SPM-1		Set ID	SI	4	1:1	R	Fixed value =1
SPM-4		Sample Type	CWE	250	1:1	R	The precise nature of the entity that is the source material for the observation (e.g. SER^serum^HL70487). See Sample types and input volume in Assays (111)
	SPM-4-1	Identifier	ST	20	1:1		Code for sample type (e.g. SER)
	SPM-4-2	Text	ST	199	0:1		Textual description of sample type (e.g. serum)
	SPM-4-3	Coding System	ID	12	1:1		99ROC or HL70487
SPM-11		Sample Role	CWE	250	1:1	R	The content of the field depends on the sample type defined in SPM-4.
	SPM-11-1	Identifier	ST	20	1:1		Supported values: - P = Patient sample - Q = Quality control sample
	SPM-11-2	Text	ST	199	0:1		
	SPM-11-3	Coding System	ID	12	1:1		Fixed value =HL70369
SPM-31		Other Specimen ID	EIP	80	0:1	O	Can be used for a secondary sample ID (e.g. 12345).
	SPM-31-1	ID	ST	80	1:1	R	Specimen barcode ID.

SPM - sample segment

SAC - specimen container segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SAC-3		Container ID	EI	80	1:1	R	
	SAC-3-1	Entity Identifier	ST	50	1:1		Specimen barcode. Copy of QPD-3 of QBP^Q11.
SAC-10		Carrier ID	EI	80	0:1	O	Copy of QPD-4 of QBP^Q11.
	SAC-10-1	Entity Identifier	ST	20	1:1		Rack ID.
SAC-11		Position in Carrier	NA	80	0:1	O	Copy of QPD-5 of QBP^Q11.
	SAC-11-1	Value 1	NM	16	1:1		Position in carrier.

☐ SAC - specimen container segment

ORC - common order segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
ORC-1		Order Control	ID	2	1:1	R	Fixed value = CA (cancel)
ORC-9		Transaction Date/Time	TS	14	1:1	R	
	ORC-9-1	Timestamp	DTM	14	1:1		Date/Time at which the order was created. String in the format YYYYMMDDhhmmss (e.g. 20180730123400)

☐ ORC - common order segment

OBR - observation request segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
OBR-2		Placer Order Number	EI	50	1:1	R	The LIS placer order number. The value is returned in the OBR-2 from the OUL message.
	OBR-2-1	Entity Identifier	ST	50	1:1		Unique host order id.
OBR-4		Universal Service Identifier	CE	250	1:1	R	Test (Assay) codes are specified in the ASAP. e.g. 42595-9^HBV^LN • See Test and targets in Assays (111)
	OBR-4-1	Identifier	ST	20	1:1		Test code (e.g. 42595-9).
	OBR-4-2	Text	ST	199	1:1		Test name (e.g. HBV).
	OBR-4-3	Coding System	ID	12	1:1		Example: LN for LOINC

■ OBR - observation request segment

TCD - test code details segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
TCD-1		Universal Service Identifier	CE	250	1:1	R	Test (Assay) codes are specified in the ASAP. e.g. 42595-9^HBV^LN • See Test and targets in Assays (111)
	TCD-1-1	Identifier	ST	20	1:1		Test code (e.g. 42595-9).
	TCD-1-2	Text	ST	199	1:1		Test name (e.g. HBV or copy of OBR-4-1).
	TCD-1-3	Coding System	ID	12	1:1		Example: LN for LOINC
TCD-8		Analyte Repeat Status	CE	250	0:1	O	Optional field the host can use to state that order is a reflex order. X800 DM only checks for identifier F and coding system HL70389. Other values are ignored.
	TCD-8-1	Identifier	ST	20	1:1		Example: F
	TCD-8-2	Text	ST	199	1:1		Example: Reflex test
	TCD-8-3	Coding System	ID	12	1:1		Fixed value =HL70389

■ TCD - test code details segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
TCD-9		Specimen Consumption Quantity	CQ	242	1:1	R	Example: 850^uL&&UCUM
	TCD-9-1	Process Input Volume	NM	16	1:1		Example: 850
	TCD-9-2	Quantity Units	CE	225	1:1		
	TCD-9-2-1	Coded Unit	ST	20	1:1		Fixed value = uL
	TCD-9-2-2	Text	ST	199	1:1		Can be empty or a copy of TCD-9-2-1.
	TCD-9-2-3	Coding System	ID	4	1:1		Fixed value = UCUM

☰ TCD - test code details segment

NTE - notes and comments segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
NTE-1		Set ID	SI	4	1:1	R	Sequence Number. May be used where multiple NTE segments are included in a message. Their numbering must be described in the application message definition. The values range from 1 to 9999. Numbering must be restarted in each group.
NTE-2		Source of Comment	ID	1	1:1	R	Fixed value = A
NTE-3		Comment Text	FT	65536	1:1	R	Reason for canceling.
NTE-4		Comment Type	CE	250	0:1	O	Ignored

☰ NTE - notes and comments segment

Example message

```
MSH|^~\&|HOST||X800_DM||20210915123330+0200||OML^O33^OML_O33|0b070615-e2ec-4463-92db-3fd53b21daa3|P|
2.5.1|||NE|AL||UNICODE UTF-8||LAB-28^IHE
SPM|1|sample123&ROCHE||UR^Urine^HL70487|||||P^HL70369
SAC|||sample123|||||S54|14
ORC|CA|||||20210915123330
OBR||3141||72828-7^CT/NG^LN
TCD|72828-7^CT/NG^LN|||||850^uL&&UCUM
```

🔗 Example OML^O33 message Test Order cancel

Related topics

- [Workflows](#)
 - [Solicited test order submission \(28\)](#)

Response to a test order submission (ORL^O34)

Message definition

Acknowledgment from x800 Data Manager ➔ LIS

Segment	Description	Usage	Cardinality	Reference
MSH	Message header	R	[1..1]	✎ MSH - message header segment (76)
MSA	Message Acknowledgment	R	[1..1]	✎ MSA - message acknowledge segment (77)
[{ERR}]	Message error		[0..*]	✎ ERR - error segment (80)
[			[0..1] ⁽¹⁾	
	--- SAMPLE start	R	[1..1]	
SPM	Sample information	R	[1..1]	✎ SPM - sample segment (78)
SAC	Container information	R	[1..1]	✎ SAC - specimen container segment (79)
{	--- ORDER start	R	[1..*] ⁽²⁾	
ORC	Common order	R	[1..1]	✎ ORC - common order segment (79)
}	--- ORDER end			
	--- SAMPLE end			
]				

✎ ORL^O34 message

(1) In case of an error (MSA-1=AE) or when no test order has been assigned to the sample in the test order (OML^O33), the response (ORL^O34) excludes the SAMPLE group (including the ORDER group)

(2) Same amount of ORDER groups as in the incoming test order (OML^O33)

MSH - message header segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-1		Field Delimiter	ST	1	1:1	R	The character specified in MSH-1 defines the field separator to be used for the rest of the message. The value is: (ASCII 124).
MSH-2		Encoding Characters	ST	4	1:1	R	The field contains the 4 delimiters in the following order: component separator, repetition separator, escape character, and subcomponent separator. The required values are: ^~\& (i.e. ASCII 94, 126, 92, and 38).
MSH-3		Sending Application	HD	227	1:1	R	The field uniquely identifies the sending application. The LIS server defines its content.
	MSH-3-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-5 of OML^O33
MSH-4		Sending Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-4. The system does not process the content of this field.
	MSH-4-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-6 of OML^O33
MSH-5		Receiving Application	HD	227	1:1	R	The field uniquely identifies the receiving application. The LIS server defines its content.
	MSH-5-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-3 of OML^O33
MSH-6		Receiving Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-6. The system does not process the content of this field.
	MSH-6-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-4 of OML^O33
MSH-7		Message Timestamp	TS	26	1:1	R	Date/time when the sending system created the message.
	MSH-7-1	Date/Time of Message Creation	DTM	19	1:1		String in the format YYYYMMDDhhmmss+-hhmm In this software version, DTM is represented in UTC with an (+/-) offset, not in local time. All other timestamps in the message shall relate to the time zone specified here. • YYYY = Year (four digits) • MM = Month (two digits) • DD = Day (two digits) • hh = Hour in UTC (two digits) • mm = Minute in UTC (two digits) • ss = Second in UTC (two digits) • +-hhmm = Time zone offset indicates whether the local time is ahead (+) or behind (-) UTC (e.g. 20120427123100+0100, indicates that UTC + 1 hour is the equivalent to the local time).
MSH-9		Message Type	MSG	15	1:1	R	Defines the type of message and trigger event Fixed value =ORL^O34^ORL_O42
	MSH-9-1	Message Code	IS	3	1:1		Fixed value =ORL
	MSH-9-2	Trigger Event	IS	3	1:1		Fixed value =O34
	MSH-9-3	Message Structure	IS	7	1:1		Fixed value =ORL_O42
MSH-10		Message Control ID	ST	36	1:1	R	A number or identifier that uniquely identifies the message. e.g. GUID

MSH - message header segment (ORL^O34)

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-11		Processing Mode	PT	3	1:1	R	
	MSH-11-1	Processing ID	ID	1	1:1		Fixed value = P (production)
MSH-12		HL7 Version ID	VID	60	1:1	R	The version of the HL7 standard used in constructing, interpreting, and validating the messages.
	MSH-12-1	HL7 Version ID	ID	5	1:1		Fixed value = 2.5.1
MSH-18		Character Set	ID	16	1:1	R	The character set used in the entire message. Fixed value = UNICODE UTF-8
MSH-21		Message Profile Identifier	EI	427	1:*	R	
	MSH-21-1	Entity ID	ST	50	1:1		Fixed value = LAB-28
	MSH-21-2	Namespace ID	IS	20	1:1		Fixed value = IHE

MSH - message header segment (ORL^O34)

MSA - message acknowledge segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSA-1		Acknowledgment Code	ID	2	1:1	R	The following codes are supported: - AA = Application Accepted - AR = Application Rejected - AE = Application Error Error handling (107)
MSA-2		Message Control ID	ST	36	1:1	R	Should be a copy of MSH-10 of OML^O33.

MSA - message acknowledge segment

SPM - sample segment

Usually a copy of OML^O33. Not needed in case of a response to a negative OML^O33 (ORC-1=DC) or an error (MSA-1=AE)!

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SPM-1		Set ID	SI	4	1:1	R	Fixed value = 1
SPM-2		Sample ID	EIP	80	0:1	O	Sample ID / tube barcode
	SPM-2-1	Placer Assigned Id		80	1:1		
	SPM-2-1-1	Entity ID	ST	50	1:1		Specimen Barcode ID (e.g. 12345&ROCHE)
	SPM-2-1-2	Namespace ID	IS	20	0:1		Fixed value = ROCHE
SPM-4		Sample Type	CWE	250	1:1	R	The precise nature of the entity that is the source material for the observation (e.g. SER^serum^HL70487). If the sample is a control, the value is NULL. See Sample types and input volume in Assays (111)
	SPM-4-1	Identifier	ST	20	1:1		Code for sample type (e.g. SER)
	SPM-4-2	Text	ST	199	0:1	O	Textual description of sample type (e.g. serum)
	SPM-4-3	Coding System	ID	12	1:1		99ROC or HL70487
SPM-11		Sample Role	CWE	250	1:1	R	The content of the field depends on the sample type defined in SPM-4.
	SPM-11-1	Identifier	ST	20	1:1		Supported values: • P = Patient sample • Q = Quality control sample
	SPM-11-2	Text	ST	199	0:1	O	
	SPM-11-3	Coding System	ID	12	1:1		e.g. HL70369
SPM-31		Other Specimen ID	EIP	80	0:1	O	Can be used for a secondary sample ID (e.g. 12345)
	SPM-31-1	ID	ST	80	1:1	R	Specimen barcode ID

SPM - sample segment

SAC - specimen container segment

Usually a copy of OML^O33. Not needed in case of a response to a negative OML^O33 (ORC-1=DC) or an error (MSA-1=AE)!

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SAC-3		Container ID	EI	80	1:1	O	Copy of SAC-3 of OML^O33.
	SAC-3-1	Entity Identifier	ST	50	1:1		Specimen barcode.
SAC-10		Carrier ID	EI	80	1:1	O	Copy of SAC-10 of OML^O33.
	SAC-10-1	Entity Identifier	ST	20	1:1		Rack ID.
SAC-11		Position in Carrier	NA	80	1:1	O	Copy of SAC-11 of OML^O33.
	SAC-11-1	Value 1	NM	16	1:1		Position in carrier.

■ SAC - specimen container segment

ORC - common order segment

Not needed in case of a response to a negative OML^O33 (ORC-1=DC) or an error (MSA-1=AE)!

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
ORC-1		Order Control	ID	2	1:1	R	Supported values, as response to NW (new order, ORC-1 in OML^O33): - OK = Order accepted - UA = Unable to Accept Supported values, as response to CA (cancel order, ORC-1 in OML^O33): - CR = Canceled as Requested - UC = Unable to Cancel
ORC-2		Placer Order Number	EI	50	1:1	R	
	ORC-2-1	Entity Identifier	ST	50	1:1		Copy of OBR-2 from OML^O33.
ORC-5		Order Status	ID	2	1:1	R	Supported values: - If ORC-1 is OK => SC (scheduled), IP (in progress) or CM (completed) - If ORC-1 is UA => CA (canceled) - If ORC-1 is CR => CA (canceled) - If ORC-1 is UC => IP (in progress) or CM (completed)

■ ORC - common order segment

ERR - error segment

Only in case MSA-1 is equal to AE, otherwise prohibited

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
ERR-2		Error Location	ERL	18	0:*	O	
	ERR-2-1	Segment ID	ST	3	1:1		Example: QPD
	ERR-2-2	Segment Sequence	NM	2	1:1		Example: 1
	ERR-2-3	Field Number	NM	2	0:1		Example: 3
	ERR-2-4	Field Repetition	NM	2	0:1		Example: 1
	ERR-2-5	Component Number	NM	2	0:1		Example: 1
	ERR-2-6	Sub-component Number	NM	2	0:1		Example: ""
ERR-3		Error Code	CWE	250	1:1	R	
	ERR-3-1	Identifier	ST	20	1:1		Value from the table HL70357. Limited to 100, 101, 102, 103, 200, 201, 202, 203, 207
	ERR-3-2	Text	ST	199	0:1		Not used
	ERR-3-3	Coding System	ID	12	1:1		Fixed value = HL70357
ERR-4		Severity	ID	1	1:1	R	Fixed value = E (error)
ERR-8		User Message	TX	250	0:1	O	Example: Barcode length less than 3 characters

ERR - error segment

Example messages

```
MSH|^~\&|X800 DM||HOST||20210915103324+0200||ORL^O34^ORL_O42|63ed2cb1-493f-4d9d-89e9-f40dbb9bd4b3|P|2.5.1|||||UNICODE UTF-8||LAB-28^IHE
MSA|AA|0b070615-e2ec-4463-92db-3fd53b21daa3
SPM|1|sample123&ROCHE||UR^Urine^HL70487|||||P^HL70369
SAC|||sample123|||||S54|14
ORC|OK|1||SC
```

Example ORL^O34 message Response to test order submission

```
MSH|^~\&|X800 DM||HOST||20210915103324+0200||ORL^O34^ORL_O42|63ed2cb1-493f-4d9d-89e9-f40dbb9bd4b3|P|2.5.1|||||UNICODE UTF-8||LAB-28^IHE
MSA|AA|0b070615-e2ec-4463-92db-3fd53b21daa3
SPM|1|sample123&ROCHE||UR^Urine^HL70487|||||P^HL70369
SAC|||sample123|||||S54|14
ORC|UA|01||CA
```

Example ORL^O34 message Test order already exists in Data Manager

 Related topics

- [Workflows](#)
 - [Solicited test order submission \(28\)](#)

Response to an unavailable test order (ORL^O34)

Message definition

Acknowledgment from x800 Data Manager ➔ LIS

Segment	Description	Usage	Cardinality	Reference
MSH	Message header	R	[1..1]	✎ MSH - message header segment (76)
MSA	Message Acknowledgment	R	[1..1]	✎ MSA - message acknowledge segment (77)
[{ERR}]	Message error		[0..*]	✎ ERR - error segment (80)

✎ ORL^O34 message

MSH - message header segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-1		Field Delimiter	ST	1	1:1	R	The character specified in MSH-1 defines the field separator to be used for the rest of the message. The value is: (ASCII 124).
MSH-2		Encoding Characters	ST	4	1:1	R	The field contains the 4 delimiters in the following order: component separator, repetition separator, escape character, and subcomponent separator. The required values are: ^~\& (i.e. ASCII 94, 126, 92, and 38).
MSH-3		Sending Application	HD	227	1:1	R	The field uniquely identifies the sending application. The LIS server defines its content.
	MSH-3-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-5 of OML^O33
MSH-4		Sending Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-4. The system does not process the content of this field.
	MSH-4-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-6 of OML^O33
MSH-5		Receiving Application	HD	227	1:1	R	The field uniquely identifies the receiving application. The LIS server defines its content.
	MSH-5-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-3 of OML^O33
MSH-6		Receiving Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-6. The system does not process the content of this field.
	MSH-6-1	Namespace ID	IS	20	1:1		Should be a copy of MSH-4 of OML^O33

✎ MSH - message header segment (ORL^O34)

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-7		Message Timestamp	TS	26	1:1	R	Date/time when the sending system created the message.
	MSH-7-1	Date/Time of Message Creation	DTM	19	1:1		String in the format YYYYMMDDhhmmss+-hhmm In this software version, DTM is represented in UTC with an (+/-) offset, not in local time. All other timestamps in the message shall relate to the time zone specified here. • YYYY = Year (four digits) • MM = Month (two digits) • DD = Day (two digits) • hh = Hour in UTC (two digits) • mm = Minute in UTC (two digits) • ss = Second in UTC (two digits) • +-hhmm = Time zone offset indicates whether the local time is ahead (+) or behind (-) UTC (e.g. 20120427123100+0100, indicates that UTC + 1 hour is the equivalent to the local time).
MSH-9		Message Type	MSG	15	1:1	R	Defines the type of message and trigger event Fixed value =ORL^O34^ORL_O42
	MSH-9-1	Message Code	IS	3	1:1		Fixed value =ORL
	MSH-9-2	Trigger Event	IS	3	1:1		Fixed value =O34
	MSH-9-3	Message Structure	IS	7	1:1		Fixed value =ORL_O42
MSH-10		Message Control ID	ST	36	1:1	R	A number or identifier that uniquely identifies the message. e.g. GUID
MSH-11		Processing Mode	PT	3	1:1	R	
	MSH-11-1	Processing ID	ID	1	1:1		Fixed value = P (production)
MSH-12		HL7 Version ID	VID	60	1:1	R	The version of the HL7 standard used in constructing, interpreting, and validating the messages.
	MSH-12-1	HL7 Version ID	ID	5	1:1		Fixed value =2.5.1
MSH-18		Character Set	ID	16	1:1	R	The character set used in the entire message. Fixed value =UNICODE UTF-8
MSH-21		Message Profile Identifier	EI	427	1:*	R	
	MSH-21-1	Entity ID	ST	50	1:1		Fixed value =LAB-28
	MSH-21-2	Namespace ID	IS	20	1:1		Fixed value =IHE

MSH - message header segment (ORL^O34)

MSA - message acknowledge segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSA-1		Acknowledgment Code	ID	2	1:1	R	The following codes are supported: - AA = Application Accepted - AR = Application Rejected - AE = Application Error
MSA-2		Message Control ID	ST	36	1:1	R	Should be a copy of MSH-10 of OML^O33.

MSA - message acknowledge segment

ERR - error segment

Only in case MSA-1 is equal to AE, otherwise prohibited

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
ERR-2		Error Location	ERL	18	0:*	O	
	ERR-2-1	Segment ID	ST	3	1:1		Example: QPD
	ERR-2-2	Segment Sequence	NM	2	1:1		Example: 1
	ERR-2-3	Field Number	NM	2	0:1		Example: 3
	ERR-2-4	Field Repetition	NM	2	0:1		Example: 1
	ERR-2-5	Component Number	NM	2	0:1		Example: 1
	ERR-2-6	Sub-component Number	NM	2	0:1		Example: ""
ERR-3		Error Code	CWE	250	1:1	R	
	ERR-3-1	Identifier	ST	20	1:1		Value from the table HL70357. Limited to 100, 101, 102, 103, 200, 201, 202, 203, 207
	ERR-3-2	Text	ST	199	0:1		Not used
	ERR-3-3	Coding System	ID	12	1:1		Fixed value = HL70357
ERR-4		Severity	ID	1	1:1	R	Fixed value = E (error)
ERR-8		User Message	TX	250	0:1	O	Example: Barcode length less than 3 characters

ERR - error segment

Example messages

```
MSH|^~\&|X800 DM||HOST||20210915103324+0200||ORL^O34^ORL_O42|63ed2cb1-493f-4d9d-89e9-f40dbb9bd4b3|P|
2.5.1|||||UNICODE UTF-8|||LAB-28^IHE
MSA|AA|0b070615-e2ec-4463-92db-3fd53b21daa3
```

 Example ORL^O34 message Response to test order submission

Related topics

- [Workflows](#)
 - [Solicited test order submission \(28\)](#)

Test result (OUL^R22)

Automatic result release

The system can be configured to send results automatically to the LIS after being calculated.

Enabling automatic result release does not affect the structure of the result message.

Message definition

Result upload from x800 Data Manager ➔ LIS.

Segment	Description	Usage	Cardinality	Reference
MSH	Message header	R	[1..1]	✎ MSH - message header segment (87)
	--- SAMPLE start	R	[1..1]	
SPM	Sample information	R	[1..1]	✎ SPM - sample segment (89)
SAC	Container information	R	[1..1]	✎ SAC - specimen container segment (90)
[INV]	INV Segement usage in the container segment: Only in case of a control result (SPM-11 = Q)	C	[0..1]	✎ INV - inventory detail segment usage in the container segment (90)
{	--- ORDER start	R	[1..*]	
OBR	Observation order	R	[1..1]	✎ OBR - observation request segment (91)
ORC	Common order	R	[1..1]	✎ ORC - common order segment (92)
[NTE]	Notes and comments	O	[0..1]	✎ NTE - notes and comments segment (92)
{	--- RESULT start	R	[1..*]	
OBX	Observation result (channel)	R	[1..1]	✎ OBX - observation result segment for target result (93)
TCD	Test code detail	R	[1..1]	✎ TCD - test code details segment (95)
[{INV}]	Inventory segment to provide information about used reagents and consumables	C	[0..*]	✎ INV - inventory detail segment to provide information about used reagents and consumables (96)
[OBX]	Supplemental OBX when ct values of target and IC/QS are sent	C	[0..1]	✎ OBX - supplemental observation result segment when ct values of target and IC/QS are sent (98)
[OBX]	Supplemental OBX for the reference to the control result	C	[0..1]	✎ OBX - Supplemental for the reference to the control result (99)

📋 OUL^R22 message

Roche Diagnostics

x800 Data Manager · Software version 2.1 and higher · Host Interface Manual · Version 5.3

Segment	Description	Usage	Cardinality	Reference
}	--- RESULT end			
}	--- ORDER end			
	--- SAMPLE end			

■ OUL^R22 message

MSH - message header segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-1		Field Delimiter	ST	1	1:1	R	The character specified in MSH-1 defines the field separator to be used for the rest of the message. The value is: (ASCII 124).
MSH-2		Encoding Characters	ST	4	1:1	R	The field contains the 4 delimiters in the following order: component separator, repetition separator, escape character, and subcomponent separator. The required values are: ^~\& (i.e. ASCII 94, 126, 92, and 38).
MSH-3		Sending Application	HD	227	1:1	R	The field uniquely identifies the sending application. The LIS server defines its content.
	MSH-3-1	Namespace ID	IS	20	1:1		Configurable in Host Settings , by default X800DM.
MSH-4		Sending Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-4. The system does not process the content of this field.
	MSH-4-1	Namespace ID	IS	20	1:1		Not used
MSH-5		Receiving Application	HD	227	1:1	R	The field uniquely identifies the receiving application. The LIS server defines its content.
	MSH-5-1	Namespace ID	IS	20	1:1		Fixed value =HOST
MSH-6		Receiving Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-6. The system does not process the content of this field.
	MSH-6-1	Namespace ID	IS	20	1:1		Not used

■ MSH - message header segment (OUL^R22)

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-7		Message Timestamp	TS	26	1:1	R	Date/time when the sending system created the message.
	MSH-7-1	Date/Time of Message Creation	DTM	19	1:1		String in the format <code>YYYYMMDDhhmmss+-hhmm</code> In this software version, DTM is represented in UTC with an (+/-) offset, not in local time. All other timestamps in the message shall relate to the time zone specified here. • <code>YYYY</code> = Year (four digits) • <code>MM</code> = Month (two digits) • <code>DD</code> = Day (two digits) • <code>hh</code> = Hour in UTC (two digits) • <code>mm</code> = Minute in UTC (two digits) • <code>ss</code> = Second in UTC (two digits) • <code>+-hhmm</code> = Time zone offset indicates whether the local time is ahead (+) or behind (-) UTC (e.g. <code>20120427123100+0100</code>, indicates that UTC + 1 hour is the equivalent to the local time).
MSH-9		Message Type	MSG	15	1:1	R	Defines the type of message and trigger event Fixed value = <code>OUL^R22^OUL_R22</code>
	MSH-9-1	Message Code	IS	3	1:1		Fixed value = <code>OUL</code>
	MSH-9-2	Trigger Event	IS	3	1:1		Fixed value = <code>R22</code>
	MSH-9-3	Message Structure	IS	7	1:1		Fixed value = <code>OUL_R22</code>
MSH-10		Message Control ID	ST	36	1:1	R	The field contains a number or identifier that uniquely identifies the message. The receiving system returns this ID to the sending system in the MSA-2 segment (ACK^R22). e.g. <code>GUID</code>
MSH-11		Processing Mode	PT	3	1:1	R	
	MSH-11-1	Processing ID	ID	1	1:1		Fixed value = <code>P</code> (production)
MSH-12		HL7 Version ID	VID	60	1:1	R	The field specifies the version of the HL7 standard used in constructing, interpreting, and validating the messages.
	MSH-12-1	HL7 Version ID	ID	5	1:1		Fixed value = <code>2.5.1</code>
MSH-15		Accept Acknowledgment Type	ID	2	1:1	R	Conditions under which accept acknowledgments are required to be returned in response to this message. Fixed value = <code>NE</code> (never)
MSH-16		Application Acknowledgment Type	ID	2	1:1	R	Conditions under which application acknowledgments are required to be returned in response to this message. Fixed value = <code>AL</code> (always)
MSH-18		Character Set	ID	16	1:1	R	The field defines the character set used in the entire message. Fixed value = <code>UNICODE UTF-8</code>
MSH-21		Message Profile Identifier	EI	427	1:*	R	
	MSH-21-1	Entity ID	ST	50	1:1		Fixed value = <code>LAB-29</code>
	MSH-21-2	Namespace ID	IS	20	1:1		Fixed value = <code>IHE</code>

MSH - message header segment (OUL^R22)

SPM - sample segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SPM-1		Set ID	SI	4	1:1	R	Fixed value = 1
SPM-2		Sample ID	EIP	80	0:1	O	Sample ID / tube barcode
	SPM-2-1	Placer Assigned Id		80	1:1		
	SPM-2-1-1	Entity ID	ST	50	1:1		Specimen Barcode ID (e.g. 12345&ROCHE)
	SPM-2-1-2	Namespace ID	IS	20	0:1		Fixed value = ROCHE
SPM-4		Sample Type	CWE	250	1:1	R	The precise nature of the entity that is the source material for the observation (e.g. SER^serum^HL70487). If the sample is a control, the value is empty (" " - two quotation marks) • See Sample types and input volume in Assays (111)
	SPM-4-1	Identifier	ST	20	1:1		Code for sample type (e.g. SER)
	SPM-4-2	Text	ST	199	0:1	O	Textual description of sample type (e.g. serum)
	SPM-4-3	Coding System	ID	12	1:1		99ROC or HL70487
SPM-11		Sample Role	CWE	250	1:1	R	The content of the field depends on the sample type defined in SPM-4.
	SPM-11-1	Identifier	ST	20	1:1		Supported values: - P = Patient sample - Q = Quality control sample
	SPM-11-2	Text	ST	199	0:1	O	
	SPM-11-3	Coding System	ID	12	1:1		e.g. HL70369
SPM-31		Other Specimen ID	EIP	80	0:1	O	Copy of SPM-31 from OML^O33. In case of a manual order on the instrument UI, the Secondary Sample ID information is added.
	SPM-31-1	ID	ST	80	1:1	R	

SPM - sample segment

SAC - specimen container segment

In case of a control (SPM-11=Q) the segment is empty but included in the message.

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
SAC-3		Container ID	EI	80	0:1	O	
	SAC-3-1	Entity Identifier	ST	50	1:1		Sample ID. Empty in case of a control result.
SAC-10		Carrier ID	EI	80	0:1	O	Copy of QPD-4 of QBP^Q11. Empty in case of a control result.
	SAC-10-1	Entity Identifier	ST	20	1:1		Rack ID. In case the sample is already unloaded when the result is reported, this field should contain the position where the sample has been placed.
SAC-11		Position in Carrier	NA	80	0:1	O	Copy of QPD-5 of QBP^Q11. Empty in case of a control result.
	SAC-11-1	Value 1	NM	16	1:1		Position in carrier.

■ SAC - specimen container segment

INV - inventory detail segment usage in the container segment

Only in case of a control result.

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
INV-1		Substance Identifier	CE	250	1:1	R	Name of the control (e.g CT/NG (+) C^^99ROC).
	INV-1-1	Identifier	ST	20	1:1	R	Accepted values: - Negative control: (-) Ctrl - Positive control: See values in table <i>Control results</i> in the assay description (e.g. CT/NG (+) C).
	INV-1-2	Text	ST	199	0:1	O	Not used
	INV-1-3	Coding System	ID	12	1:1	R	Fixed value = 99ROC

■ INV - inventory detail segment usage in the container segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
INV-2		Substance Status	CE	250	1:1	R	
	INV-2-1	Identifier	ST	20	1:1	R	Fixed value =OK
	INV-2-2	Text	ST	199	0:1	O	Not used
	INV-2-3	Coding System	ID	12	1:1	R	Fixed value =HL70383
INV-3		Substance Type	CE	250	1:1	R	
	INV-3-1	Identifier	ST	20	1:1	R	Fixed value =CO
	INV-3-2	Text	ST	199	0:1	O	Not used
	INV-3-3	Coding System	ID	12	1:1	R	Fixed value =HL70384
INV-12		Expiration Date/Time	TS	14	0:1	O	String in the format YYYYMMDDHHMMSS Time zone indicator is not supported, and is assumed to be the same as the value in MSH-7 Date/Time of Message.
INV-16		Manufacturer Lot Number	ST	200	0:1	O	

INV - inventory detail segment usage in the container segment

OBR - observation request segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
OBR-2		Placer Order Number	EI	50	0:1	C	Copy of OBR-2 from OML^O33. Empty if: - The order has been manually created on the x800 Data Manager - A reflex order has been requested by the Analyzer (i.e. manually on the instrument user interface)
	OBR-2-1	Entity Identifier	ST	50	1:1		Unique host order id.
OBR-4		Universal Service Identifier	CE	250	1:1	R	Test (Assay) codes are specified in the ASAP. e.g. 42595-9^HBV^LN See Test and targets in Assays (111)
	OBR-4-1	Identifier	ST	20	1:1		Test code (e.g. 42595-9).
	OBR-4-2	Text	ST	199	1:1		Test name (e.g. HBV).
	OBR-4-3	Coding System	ID	12	1:1		Example: LN (LOINC)
OBR-11		Specimen Action Code	ID	1	0:1	O	Fixed value =G when the result was obtained in a reflex test order requested by the Analyzer (i.e. manually on the instrument user interface). Otherwise prohibited.

OBR - observation request segment

ORC - common order segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
ORC-1		Order Control	ID	2	1:1	R	Fixed value = SC (status change)
ORC-5		Order Status	ID	2	1:1	R	Fixed value = CM (order is completed)
ORC-8		Parent Order Number	EIP	50	0:*	O	Placer order number(s) in case the reflex has been requested by the Analyzer (i.e. manually on the instrument user interface). Therefore no reflex order from the LIS has been received and the value refers to the initial LIS test-order e.g. 4~5~6 In the situation where the reflex has been sent by the LIS, the value is empty. Otherwise prohibited.
	ORC-8-1	Placer Assigned Identifier	EI	50	1:1		
	ORC-8-1-1	Entity Identifier	ST	50	1:1	R	Copy of OBR-2 from parent OML^O33 message.

ORC - common order segment

NTE - notes and comments segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
NTE-1		Set ID	SI	4	1:1	R	Sequence number.
NTE-2		Source of Comment	ID	1	1:1	R	Fixed value = Z (analyzer)
NTE-3		Comment Text	FT	65536	1:1	R	Comment source: - Has been contained in the OML^O33 from the host - Has been added by the user at manual order creation - Has been added/modified at result release

NTE - notes and comments segment

OBX - observation result segment for target result

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
OBX-1		Set ID	SI	4	1:1	R	Sequence number.
OBX-2		Value Type	ID	2	0:1	RE	The field contains the format of the observation value in OBX-5. Supported values: - NM = numeric (OBX-5 contains a numeric result) - ST = string - Empty = result is not available or invalid
OBX-3		Observation Identifier	CE	250	1:1	R	Target (channel) result code. Target (channel) result codes are specified in the ASAP. ☞ See ASAP definition in Glossary (207)
	OBX-3-1	Identifier	ST	20	1:1		Test code (e.g. NG)
	OBX-3-2	Test	ST	199	1:1		Test name or copy of OBX-1 (e.g. NG)
	OBX-3-3	Coding System	ID	12	1:1		Fixed value = 99ROC
OBX-4		Observation Sub-ID	OG	20	1:1	R	
	OBX-4-1	Run number	ST	20	1:1	R	Fixed value = 1
OBX-5		Observation Value	ST/ NM	99999	1:*	RE	Supported data types: - ST for qualitative assay - NM for quantitative assay Supported values for channel results: - For numeric results, the value contains the 3 significant digits of the value represented with precision of 3. The multiplier factor of OBX-6 must be considered for the result calculation. - Empty = result is not available or invalid ☞ For assay-specific codes, see Assays (111)

☞ OBX - observation result segment for target result

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
OBX-6		Units	CE	250	0:1	C	Empty if no result available (e.g. invalid result) and empty for qualitative assays. The field defines base 10 raised to the power of the exponent together with the unit and the coding system. To calculate the result, OBX-6 must be multiplied with the significant digits of OBX-5, for example: - OBX-5 = 532 - OBX-6 = $10^{*2} \cdot [\text{IU}] / \text{mL}^{\wedge\wedge\text{UCUM}}$ - Result = $532 * 10^2 = 53200 [\text{IU}] / \text{mL}$ The ASAP allows the following 2 options: $10^{*}\text{exponent} \cdot [\text{IU}] / \text{mL}^{\wedge\wedge\text{UCUM}}$ $10^{*}\text{exponent} \cdot \{\text{copies}\} / \text{mL}^{\wedge\wedge\text{UCUM}}$
OBX-6-1	Identifier	ST	ST	20	1:1		The ASAP supports the following 2 values: $10^{*}\text{exponent} \cdot [\text{IU}] / \text{mL}$ $10^{*}\text{exponent} \cdot \{\text{copies}\} / \text{mL}$ The number 10 is the base whereas the exponent is a positive or negative integer value depending on the test result, for example: $10^{*2} \cdot [\text{IU}] / \text{mL} \rightarrow 10^2 \rightarrow 100[\text{IU}] / \text{mL}$ $10^{*-1} \cdot [\text{IU}] / \text{mL} \rightarrow 10^{-1} \rightarrow 0.1[\text{IU}] / \text{mL}$
OBX-6-2	Text	ST	ST	199	1:1		Not used
OBX-6-3	Coding System	ID	ID	12	1:1		Fixed value = UCUM
OBX-8		Interpretation Codes	CWE	5	1:*	R	The field is used in combination with OBX-11 to notify the status of the results. One or more code or error flags can be used. Supported values are dynamic and depend on the implemented ASAP. - For current values, see Assays (111) - For supported error flags for invalid results, see Error flags (197)
OBX-8-1	Identifier	ST	ST	20	1:1		Interpretation result (e.g. NR, RR, VAL, etc.). In case of an invalid result, this field contains the flag code. Flags on observation level need to be distributed to each target (OBX). The error flag U04 is used for canceling a test-order. Several repetitions are possible e.g. C01T^99ROC~P10P^99ROC~P15^99ROC
OBX-8-2	Text	ST	ST	199	0:1		Empty "" or flag description.
OBX-8-3	Coding System	ID	ID	12	1:1		Fixed value = 99ROC
OBX-11		Observation Result Status	ID	1	1:1	R	The field is used in combination with OBX-8 to notify the status of the results. The system supports the following values: - F = Final and valid result - X = No results can be obtained for this observation, or order was canceled

OBX - observation result segment for target result

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
OBX-16		Responsible Observer	ST	80	1:1	R	The user that released the result. In automatically released results, the system sends the Data Manager ID (X800DMSYSTEM) rather than the user.
OBX-18		Equipment Instance Identifier	EI	427	2:3	R	The field identifies the Equipment Instance (e.g. Analyzer, Analyzer module, group of Analyzers, etc.) responsible for the production of the observation. 1st repetition: Instrument model ID ^ Manufacturer ID 2nd repetition: Instrument serial number ^ Manufacturer ID 3rd repetition (optional): vendor-specific value (not implemented) The tilde (~) is the HL7 repetition delimiter. Example: c5800^Roche~c5800.1234^Roche
	OBX-18-1	Instrument Model ID	ST	50	1:1		c5800, c6800, or c8800.
	OBX-18-2	Manufacturer	IS	20	1:1		Fixed value = Roche
OBX-19		Date/Time of the Analysis	TS	14	1:1	R	The field is used to transfer the time stamp (UTC time) associated with the generation of the analytical result by the instrument specified in OBX-18. The format is YYYYMMDDHHMMSS
OBX-21		Observation Instance identifier	EI	50	0:1	O	
	OBX-21-1	Entity Identifier	ST	50	1:1		Run ID in which the result has been produced.
OBX-29		Observation Type	ID	4	1:1	R	Fixed value = RSLT (result)

OBX - observation result segment for target result

TCD - test code details segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
TCD-1		Universal Service Identifier	CE	250	1:1	R	Test (Assay) codes are specified in the ASAP. e.g. 42595-9^HBV^LN See Test and targets in Assays (111)
	TCD-1-1	Identifier	ST	20	1:1		Test code (e.g. 42595-9).
	TCD-1-2	Text	ST	199	1:1		Test name (e.g. HBV).
	TCD-1-3	Coding System	ID	12	1:1		Example: LN for LOINC

TCD - test code details segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
TCD-9		Specimen Consumption Quantity	CQ	242	1:1	RE	Empty for control results. Example: 850^uL&&UCUM
	TCD-9-1	Process Input Volume	NM	16	1:1		Example: 850
	TCD-9-2	Quantity Units	CE	225	1:1		
	TCD-9-2-1	Coded Unit	ST	20	1:1		Fixed value =uL
	TCD-9-2-2	Text	ST	199	1:1		Can be empty or a copy of TCD-9-2-1.
	TCD-9-2-3	Coding System	ID	4	1:1		Fixed value =UCUM

☐ TCD - test code details segment

INV - inventory detail segment to provide information about used reagents and consumables

Inventory segment to provide information about used reagents and consumables. Since target results could have been processed in different runs, the INV segments need to be sent for each target OBX. In case no sample tracking information is available (order cancel), no INV segment is sent.

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
INV-1		Substance Identifier	CE	250	1:1	R	
	INV-1-1	Identifier	ST	20	1:1	R	Name of the reagent or consumable (e.g. MGP cassette^^99ROC). Accepted values: • Reagent: Reagent cassette • MGP Cassette: MGP cassette • Lysis reagent: Lysis reagent • Diluent: Diluent • Wash reagent: Wash reagent • Tip rack: Tip rack • Amplification plate: Amplification plate • Processing plate: Processing plate
	INV-1-2	Text	ST	199	0:1	O	Empty ""
	INV-1-3	Coding System	ID	12	1:1	R	Fixed value =99ROC

☐ INV - inventory detail segment to provide information about used reagents and consumables

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
INV-2		Substance Status	CE	250	1:1	R	
	INV-2-1	Identifier	ST	20	1:1	R	Fixed value =OK
	INV-2-2	Text	ST	199	0:1	O	Not used
	INV-2-3	Coding System	ID	12	1:1	R	Fixed value =HL70383
INV-3		Substance Type	CE	250	1:1	R	
	INV-3-1	Identifier	ST	20	1:1	R	Supported values: - MR = Multiple test reagent - DI = Diluent - CO = Control - LI = Measurable liquid item - SC = Countable solid item
	INV-3-2	Text	ST	199	0:1	O	Not used
	INV-3-3	Coding System	ID	12	1:1	R	Fixed value =HL70384
INV-12		Expiration Date/Time	TS	14	0:1	O	String in the format YYYYMMDDHHMMSS Time zone indicator is not supported, and is assumed to be the same as the value in MSH-7 Date/Time of Message.
INV-16		Manufacturer Lot Number	ST	200	0:1	O	

INV - inventory detail segment to provide information about used reagents and consumables

Related topics

- Workflows
 - Result upload (28)
 - Test order delete notification (28)

OBX - supplemental observation result segment when ct values of target and IC/QS are sent

In case of an invalid result or order cancel, no supplemental OBX shall be sent.

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
OBX-1		Set ID	SI	4	1:1	R	The field contains the sequence number of the OBX element in the message. • Values range from 1 to 9999
OBX-2		Value Type	ID	2	1:1	R	The field contains the format of the observation value in OBX-5. Fixed value = NA (Numeric Array)
OBX-3		Observation Identifier	CE	250	1:1	R	Target (channel) result code. Target (channel) result codes are specified in the ASAP.
	OBX-3-1	Identifier	ST	20	1:1		Copy from main OBX-3-1.
	OBX-3-2	Test	ST	199	1:1		Copy from main OBX-3-2.
	OBX-3-3	Coding System	ID	12	1:1		Copy from main OBX-3-3.
	OBX-3-4	Alternate Identifier	ST	20	0:1		Fixed value = S_OTHER
	OBX-3-5	Alternate Text	ST	199	0:1		Fixed value = Other Supplemental
	OBX-3-6	Alternate Coding System	ID	12	0:1		Fixed value = IHELAW
OBX-4		Observation Sub-ID	OG	20	0:1	C	
	OBX-4-1		ST	20	1:1	R	Fixed value = 1
OBX-5		Observation Value	NA	99999	1:*	R	Result value. Supplemental OBX is sent with each target OBX. The target ct value will be different in each supplemental OBX while the IC/CT values remain the same.
	OBX-5-1	Target Ct Value	NM	16	1:1	RE	Empty if target result is non reactive. Number with optional decimal point. No scientific notation. E.g. 27.76
	OBX-5-2	IC Ct Value	NM	16	1:1	RE	Empty if quantitative target. Number with optional decimal point. No scientific notation. E.g. 27.76
	OBX-5-3	QS Ct Value	NM	16	1:1	RE	Empty if qualitative target. Number with optional decimal point. No scientific notation. E.g. 27.76
OBX-6		Units	CE	250	1:1	C	Not used
OBX-8		Interpretation Codes	CWE	5	1:*	R	Copy of the OBX-8 field from the observation target result segment.
OBX-11		Observation Result Status	ID	1	1:1	R	Copy of the OBX-11 field from the observation target result segment.

OBX - supplemental observation result segment when ct values of IC/QS are sent

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
OBX-16		Responsible Observer	ST	80	1:1	R	Copy of the OBX-16 field from the observation target result segment.
OBX-18		Equipment Instance Identifier	EI	427	2:*	R	Copy of the OBX-18 field from the observation target result segment.
OBX-19		Date/Time of the Analysis	TS	14	1:1	R	Copy of the OBX-19 field from the observation target result segment.
OBX-21		Observation Instance identifier	EI	50	0:1	O	Copy of the OBX-21 field from the observation target result segment.
OBX-29		Observation Type	ID	4	1:1	R	Fixed value = RSLT (result)

OBX - supplemental observation result segment when ct values of IC/QS are sent

OBX - Supplemental for the reference to the control result

In case of an invalid result, this supplemental OBX shall be still sent.

In case of order cancel or control result, this supplemental OBX is not sent.

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
OBX-1		Set ID	SI	4	1:1	R	The field contains the sequence number of the OBX element in the message. • Values range from 1 to 9999
OBX-2		Value Type	ID	2	1:1	R	Fixed value = EI (entity identifier)
OBX-3		Observation Identifier	CE	250	1:1	R	Target (channel) result code.
	OBX-3-1	Identifier	ST	20	1:1		Fixed value = QCTID
	OBX-3-2	Test	ST	199	1:1		Fixed value = QC Test ID
	OBX-3-3	Coding System	ID	12	1:1		Fixed value = 99ROC
	OBX-3-4	Alternate Identifier	ST	20	0:1		Fixed value = S_OTHER
	OBX-3-5	Alternate Text	ST	199	0:1		Fixed value = Other Supplemental
	OBX-3-6	Alternate Coding System	ID	12	0:1		Fixed value = IHELAW

OBX - observation result segment for the reference to the control result

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
OBX-4		Observation Sub-ID	OG	20	0:1	C	
	OBX-4-1		ST	20	1:1	R	Fixed value = 1
OBX-5		Observation Value	ST	99999	1:*	R	List of control ID's, separated by the repeat character ~ Reference to the control ID's. E.g. the patient result is related to a positive and negative control, two values are listed: <positive control ID>~<negative control ID> ❏ See Assays (111)
OBX-6		Units	CE	250			Not used
OBX-8		Interpretation Codes	CWE	5	1:*	R	Copy of the OBX-8 field from the observation target result segment.
OBX-11		Observation Result Status	ID	1	1:1	R	Copy of the OBX-11 field from the observation target result segment.
OBX-16		Responsible Observer	ST	80	1:1	R	Copy of the OBX-16 field from the observation target result segment.
OBX-18		Equipment Instance Identifier	EI	427	2:*	R	Copy of the OBX-18 field from the observation target result segment.
OBX-19		Date/Time of the Analysis	TS	14	1:1	R	The field is used to transfer the time stamp (UTC time) associated with the generation of the analytical result by the instrument specified in OBX-18. The format is YYYYMMDDHHMMSS
OBX-21		Observation Instance identifier	EI	50	0:1	O	Copy of the OBX-21 field from the observation target result segment.
OBX-29		Observation Type	ID	4	1:1	R	Fixed value = RSLT (result)
❏ OBX - observation result segment for the reference to the control result							

Cancel (OUL^R22)

Order canceled is handled the same way as an invalid result. OBX-11 = X indicates the non availability of a result.

•  Test result (OUL^R22) (86)

Reference to IHE LAW Definition

Should a specimen be damaged or lost, the Analyzer Manager will suspend or cancel its Analytical Work Order until the replacement specimen arrives.

OBX-11 "Observation Result Status" is the status of an observation produced by a run of the AWOS. Currently only F and X are supported. It is interpreted as follows:

- F represents a preliminary result in a progression of results leading to the reportable result (such as the status of cultures in Microbiology).
- R (entered, not verified) is used when the Analyzer has performed multiple runs for an AWOS, has selected the best run whose result is flagged F, and is also reporting the results produced by the other runs for traceability. The results of these other runs are flagged R to inform the Analyzer Manager that these are not candidates to be reportable. The only candidate result is the one flagged F.
- F marks a final result of an Observation Run. The result is a candidate to be reported to the ordering provider. In case of multiple runs for an AWOS, if the results of all runs are flagged F, all of them are candidates and the decision to choose the best one is left up to the Analyzer Manager. Conversely, if the results of all runs are flagged R but one flagged F, this one is the only candidate.
- C represents a correction of a result previously sent as final. The correction stays in the same Observation Run.
- X means that the Analyzer was not able to produce any result for this observation in this Observation Run. There will be no results for the observation in this run. If this is the only run, this observation will have no result.

 Related topics

- [Workflows](#)
 - [Test order delete notification \(28\)](#)

Result Accepted (ACK^R22)

Message definition

Acknowledgment from LIS x800 Data Manager

Segment	Description	Usage	Cardinality	Reference
MSH	Message header	R	[1..1]	MSH - message header segment (103)
MSA	Query parameter definition	R	[1..1]	MSA - message acknowledge segment (105)
[{ERR}]	Message error	C	[0..*]	ERR - error segment (105)

ACK^R22 message

MSH - message header segment

Field	Component		Data Type	Maximum Length	Cardinality	Optional	Description
	Element Name						
MSH-1	Field Delimiter	ST	1	1:1	R	The character specified in MSH-1 defines the field separator to be used for the rest of the message. The value is: (ASCII 124).	
MSH-2	Encoding Characters	ST	4	1:1	R	The field contains the 4 delimiters in the following order: component separator, repetition separator, escape character, and subcomponent separator. The required values are: ^~\& (i.e. ASCII 94, 126, 92, and 38).	
MSH-3	Sending Application	HD	227	1:1	R	The field uniquely identifies the sending application. The LIS server defines its content.	
	MSH-3-1	Namespace ID	IS	20	1:1	Should be a copy of MSH-5 of OUL^R22.	
MSH-4	Sending Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-4. The system does not process the content of this field.	
	MSH-4-1	Namespace ID	IS	20	1:1	Should be a copy of MSH-6 of OUL^R22.	
MSH-5	Receiving Application	HD	227	1:1	O	The field uniquely identifies the receiving application. The LIS server defines its content.	
	MSH-5-1	Namespace ID	IS	20	1:1	Should be a copy of MSH-3 of OUL^R22.	
MSH-6	Receiving Facility	HD	227	0:1	O	HL7 does not specify any values for MSH-6. The system does not process the content of this field.	
	MSH-6-1	Namespace ID	IS	20	1:1	Should be a copy of MSH-4 of OUL^R22.	

MSH - message header segment (ACK^R22)

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSH-7		Message Timestamp	TS	26	1:1	R	Date/time when the sending system created the message.
	MSH-7-1	Date/Time of Message Creation	DTM	19	1:1		String in the format YYYYMMDDhhmmss+-hhmm In this software version, DTM is represented in UTC with an (+/-) offset, not in local time. All other timestamps in the message shall relate to the time zone specified here. • YYYY = Year (four digits) • MM = Month (two digits) • DD = Day (two digits) • hh = Hour in UTC (two digits) • mm = Minute in UTC (two digits) • ss = Second in UTC (two digits) • +-hhmm = Time zone offset indicates whether the local time is ahead (+) or behind (-) UTC (e.g. 20120427123100+0100, indicates that UTC + 1 hour is the equivalent to the local time).
MSH-9		Message Type	MSG	15	1:1	R	Defines the type of message and trigger event Fixed value =ACK^R22^ACK
	MSH-9-1	Message Code	IS	3	1:1		Fixed value =ACK
	MSH-9-2	Trigger Event	IS	3	1:1		Fixed value =R22
	MSH-9-3	Message Structure	IS	7	1:1		Fixed value =ACK
MSH-10		Message Control ID	ST	36	1:1	R	The field contains a number or identifier that uniquely identifies the message (e.g. GUID).
MSH-11		Processing Mode	PT	3	1:1	R	
	MSH-11-1	Processing ID	ID	1	1:1		Fixed value =P (production)
MSH-12		HL7 Version ID	VID	60	1:1	R	The field specifies the version of the HL7 standard used in constructing, interpreting, and validating the messages.
	MSH-12-1	HL7 Version ID	ID	5	1:1		Fixed value =2.5.1
MSH-18		Character Set	ID	16	1:1	R	The field defines the character set used in the entire message. Fixed value =UNICODE UTF-8
MSH-21		Message Profile Identifier	EI	427	1:*	R	
	MSH-21-1	Entity ID	ST	50	1:1		Fixed value =LAB-29
	MSH-21-2	Namespace ID	IS	20	1:1		Fixed value =IHE

MSH - message header segment (ACK^R22)

MSA - message acknowledge segment

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
MSA-1		Acknowledgment Code	ID	2	1:1	R	The following codes are supported: - AA = Application Accepted - AR = Application Rejected (also applied in case of specimen, container, or identification) - AE = Application Error - Error handling (107)
MSA-2		Message Control ID	ST	50	1:1	R	Should be a copy of MSH-10 of OUL^R22.

MSA - message acknowledge segment

ERR - error segment

Error segment if message acknowledge is negative (MSA-1 is unequal AA).

Field	Component	Element Name	Data Type	Maximum Length	Cardinality	Optional	Description
ERR-2		Error Location	ERL	18	0:*	O	
	ERR-2-1	Segment ID	ST	3	1:1		Example: OBX
	ERR-2-2	Segment Sequence	NM	2	1:1		Example: 1
	ERR-2-3	Field Number	NM	2	0:1		Example: 2
	ERR-2-4	Field Repetition	NM	2	0:1		Example: 1
	ERR-2-5	Component Number	NM	2	0:1		Example: 1
	ERR-2-6	Sub-component Number	NM	2	0:1		Example: ""
ERR-3		Error Code	CWE	250	1:1	R	
	ERR-3-1	Identifier	ST	20	1:1		Value from the table HL70357. Limited to 100, 101, 102, 103, 200, 201, 202, 203, 207
	ERR-3-2	Text	ST	199	0:1		Not used
	ERR-3-3	Coding System	ID	12	1:1		Fixed value = HL70357
ERR-4		Severity	ID	1	1:1	R	Fixed value = E (error)
ERR-8		User Message	TX	250	0:1	O	Example: Unknown value type

ERR - error segment

 Related topics

- [Workflow](#)
 - [Result upload \(28\)](#)
 - [Test order delete notification \(28\)](#)

Error handling

This chapter describes how the software handles communication errors.

In this chapter

4

Error handling 107

Error handling

The software provides a simple error handling as described below.

Message rejection by the LIS

If the LIS does not accept the test order query message (QBP^Q11), it sends an Application Rejected (AR) or Application Error (AE) in RSP^K11 MSA-1 in return to the software.

The software discards the message and informs the user about the rejected message. The software remains connected to the LIS.

• Application codes for message rejection (108)

No reply from the LIS

If the software sends a message to the LIS and receives no acknowledgment within the timeout-period (configurable from 30 to 90 seconds), it resends the message another 2 times.

If the software receives no acknowledgment after sending the same message 3 times, it discards the message and notifies the user about the failed message transmission.

When the software does not receive any acknowledge after sending the same message 3 times, the software remains connected to the LIS.

The number of attempts to resend a message is limited to 2, which is not configurable.

Message rejection by the software

If the software does not accept the test order message (OML^O33) or the test result message (OUL^R22), it sends an Application Rejected (AR) or Application Error (AE) in ORL^O34 or ACK^R22 MSA-1 in return to the host.

The software remains connected to the LIS.

The software can reject a message due to the following reasons:

- The message cannot be processed because it is incomplete or contains invalid values.
- The message contains a test order that cannot be created in the software due to restrictions (e.g. reuse period time for sample IDs).
- Application codes for message rejection (108)

Application codes for message rejection

When rejecting a message, the software or LIS sends one of the following application codes in ORL^O34 or ACK^R22 MSA-1:

- Application Rejected (AR): The query message can not be processed, or values in MSH-9, MSH-11, MSH-12 or MSH-21 are not set correctly.
- Application Error (AE): Any other error in the message or an internal system error.

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Assays

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Introduction

 Assay definitions are case-sensitive! Always use the exact spelling for test and target names and sample type names as it is indicated in this chapter.

Blood screening

This section summarizes the available assays and provides assay-specific codes and specifications.

Blood screening assays listed inside this section are designed for Individual Donor Testing (IDT) and do not support pooling workflows.

In this section

- HEV - LIS mapping (117)
- Malaria - LIS mapping (119)
- MPX-E - LIS mapping (121)
- MPX - LIS mapping (123)
- WNV - LIS mapping (125)
- DPX - LIS mapping (127)
- Babesia - LIS mapping (129)
- ZIKA - LIS mapping (131)
- CHIKV/DENV - LIS mapping (133)

HEV - LIS mapping

HEV is a qualitative single-channel blood screening assay. For each test order, the software returns a single result group in the OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	850

HEV sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HEV	74858-2^HEV^LN

HEV tests

Target	Observation Identifier (OBX-3)
HEV	HEV^HEV^99ROC

HEV targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HEV	Reactive	RR
	Non-Reactive	NR
	Not used	Error flag ⁽¹⁾

HEV sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
74858-2^HEV^LN	HEV (+) C	HEV^HEV^99ROC	Valid	VAL
	(-) C		Not used	Error flag ⁽¹⁾

 HEV control results
(1) See Error flags (197).

Malaria - LIS mapping

Malaria a qualitative single-channel blood screening assay. For each test order, the software returns a single result group in the OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
Whole Blood	BLD^Whole Blood^HL70487	500

☒ Malaria sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
Malaria	50687-3^Malaria^LN

☒ Malaria tests

Target	Observation Identifier (OBX-3)
Malaria	Malaria^Malaria^99ROC

☒ Malaria targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
Malaria	Reactive	RR
	Non-Reactive	NR
	Not used	Error flag ⁽¹⁾

☒ Malaria sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
50687-3^Malaria^LN	Malaria (+) C	Malaria^Malaria	Valid	VAL
	(-) C	^99ROC	Not used	Error flag ⁽¹⁾

 Malaria control results

(1) See Error flags (197).

MPX-E - LIS mapping

MPX-E is a qualitative multi-channel blood screening assay. For each test order, the software returns a single CHANNEL RESULT group (including OBX, supplemental and INV segments) in the OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	850 / 150
SERUM	SER^serum^HL70487	850 / 150
CADAVERIC PLASMA	CP^cadavericPlasma^99ROC	850 / 150
CADAVERIC SERUM	CS^cadavericSerum^99ROC	850 / 150

MPX-E sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
MPX-E	103565-8^MPX-E^LN
MPXE-HIV/HCV/HBV	74856-6^MPXE-HIV/HCV/HBV^LN
MPXE-HBV	29610-3^MPXE-HBV^LN
MPXE-HIV	85368-9^MPXE-HIV^LN
MPXE-HEV	74858-2^MPXE-HEV^LN
MPXE-HCV	11259-9^MPXE-HCV^LN

MPX-E tests

Target	Observation Identifier (OBX-3)
HBV	HBV^HBV^99ROC
HIV	HIV^HIV^99ROC
HEV	HEV^HEV^99ROC
HCV	HCV^HCV^99ROC

MPX-E targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HBV		
HIV	Reactive	RR
HEV	Non-Reactive	NR
HCV	Not used	Error flag ⁽¹⁾

MPX-E sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
103565-8^MPX-E^LN	MPX-E (+) C	HBV^HBV^99ROC		
	(-) C	HIV^HIV^99ROC	Valid	VAL
		HEV^HEV^99ROC	Not used	Error flag ⁽¹⁾
		HCV^HCV^99ROC		

MPX-E control results

(1) See Error flags (197).

MPX - LIS mapping

MPX is a qualitative multi-channel blood screening assay. For each test order, the software returns a single CHANNEL RESULT group (including OBX, supplemental and INV segments) in the OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	850 / 150 ⁽¹⁾
SERUM	SER^serum^HL70487	850
CADAVERIC PLASMA	CP^cadavericPlasma^99ROC	150
CADAVERIC SERUM	CS^cadavericSerum^99ROC	150

MPX sample types and input volume

(1) Only for the x800-ASAP

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
MPX	74856-6^MPX^LN

MPX tests

Target	Observation Identifier (OBX-3)
HIV	HIV^HIV^99ROC
HBV	HBV^HBV^99ROC
HCV	HCV^HCV^99ROC

MPX targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HIV	Reactive	RR
HBV	Non-Reactive	NR
HCV	Not used	Error flag ⁽¹⁾

MPX sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
74856-6^MPX^LN	MPX M (+) C			
	MPX O (+) C	HIV^HIV^99ROC	Valid	VAL
	MPX 2 (+) C	HBV^HBV^99ROC	Not used	Error flag ⁽¹⁾
	(-) C	HCV^HCV^99ROC		

 MPX control results
(1) See Error flags (197).

WNV - LIS mapping

WNV is a qualitative single-channel blood screening assay. For each test order, the software returns a single result group in the OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	850 / 150 ⁽¹⁾
CADAVERIC PLASMA	CP^cadavericPlasma^99ROC	150
CADAVERIC SERUM	CS^cadavericSerum^99ROC	150

WNV sample types and input volume

(1) Only for the x800-ASAP

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
WNV	74857-4^WNV^LN

WNV tests

Target	Observation Identifier (OBX-3)
WNV	WNV^WNV^99ROC

WNV targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
WNV	Reactive	RR
	Non-Reactive	NR
	Not used	Error flag ⁽¹⁾

WNV sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
74857-4^WNV^LN	WNV (+) C	WNV^WNV^99ROC	Valid	VAL
	(-) C		Not used	Error flag ⁽¹⁾

WNV control results

(1) See Error flags (197).

DPX - LIS mapping

DPX is a qualitative and quantitative multi-channel blood screening assay. For each test order, the software returns a single result group in the OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	850

DPX sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
DPX	75540-5^DPX^LN
B19	75541-3^DPX-B19^LN

DPX tests

Target	Observation Identifier (OBX-3)
HAV	HAV^HAV^99ROC
B19	B19^B19^99ROC

DPX targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HAV	Reactive	RR
	Non-Reactive	NR
	Not used	Error flag ⁽¹⁾
B19	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

DPX sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
All	DPX H (+) C	B19^B19^99ROC	Titer (value) Not used	VAL Error flag ⁽¹⁾
		B19^B19^99ROC	Titer (value) Not used	
	DPX D (+) C	HAV^HAV^99ROC	Valid Not used	
		B19^B19^99ROC	Valid Not used	
	(-) Ctrl	HAV^HAV^99ROC	Valid Not used	
		B19^B19^99ROC	Valid Not used	

 DPX control results
(1) See Error flags (197).

Babesia - LIS mapping

Babesia is a qualitative single-channel blood screening assay. For each test order, the software returns a single result group in the OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
Whole blood	BLD^Whole Blood^HL70487	500

☰ Babesia sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
Babesia	89342-0^Babesia^LN

☰ Babesia tests

Target	Observation Identifier (OBX-3)
Babesia	Babesia^Babesia^99ROC

☰ Babesia targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
Babesia	Reactive	RR
	Non-Reactive	NR
	Not used	Error flag ⁽¹⁾

☰ Babesia sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
89342-0^Babesia^LN	Babesia (+) C	Babesia^Babesia^99ROC	Valid	VAL
	(-) C		Not used	Error flag ⁽¹⁾

 Babesia control results

(1) See Error flags (197).

ZIKA - LIS mapping

ZIKA is a qualitative single-channel blood screening assay. For each test order, the software returns a single result group in the OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	850

📖 ZIKA sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
ZIKA	86190-6^ZIKA^LN

📖 ZIKA tests

Target	Observation Identifier (OBX-3)
ZIKA	ZIKA^ZIKA^99ROC

📖 ZIKA targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
ZIKA	Reactive	RR
	Non-Reactive	NR
	Not used	Error flag ⁽¹⁾

📖 ZIKA sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
86190-6^ZIKA^LN	ZIKA (+) C	-ZIKA^ZIKA^99ROC	Valid	VAL
	(-) C		Not used	Error flag ⁽¹⁾

 ZIKA control results

(1) See Error flags (197).

CHIKV/DENV - LIS mapping

CHIKV/DENV is a qualitative multi-channel blood screening assay. For each test order, the software returns a single result group in the OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	850

CHIKV/DENV sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
CHIKV/DENV	85582-5^CHIKV-DENV^LN
CHIKV	85583-3^CHIKV^LN
DENV	85584-1^DENV^LN

CHIKV/DENV tests

Target	Observation Identifier (OBX-3)
CHIKV	CHIKV^CHIKV^99ROC
DENV	DENV^DENV^99ROC

CHIKV/DENV targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
All	Reactive	RR
	Non-Reactive	NR
	Not used	Error flag ⁽¹⁾

CHIKV/DENV sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
All	CHIKV/DENV (+) C	CHIKV^CHIKV^99ROC	Valid	VAL
	(-) C	DENV^DENV^99ROC	Not used	Error flag ⁽¹⁾

 CHIKV/DENV control results

(1) See Error flags (197).

Infectious disease

This section lists the specific order, sample, and result codes used for each test.

In this section

BKV - LIS mapping (136)
CMV - LIS mapping (138)
EBV - LIS mapping (140)
HBV - LIS mapping (142)
HBV-RNA-RUO - LIS mapping (144)
HCV - LIS mapping (146)
HCV-PSC - LIS mapping (148)
HDV - LIS mapping (150)
HDV-RUO - LIS mapping (152)
HIV-1 - LIS mapping (154)
HIV-1/2-Qual-DBS - LIS mapping (156)
HIV-1/2-Qual-Ser/Pla - LIS mapping (158)
HIV-PSC - LIS mapping (160)
RESP-4FLEX - LIS mapping (162)
RESP-FLEX - LIS mapping (164)
SARS-CoV-2 Duo - LIS mapping (166)
SCoV2-FluA/B - LIS mapping (168)
SCoV2-QL - LIS mapping (170)

BKV - LIS mapping

BKV is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	200
URINE	UR^Urine^HL70487	400

☰ BKV sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
BKV	32284-2^BKV^LN

☰ BKV tests

Target	Observation Identifier (OBX-3)
BKV	BKV^BKV^99ROC

☰ BKV targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
BKV	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

☰ BKV sample result codes for OBX segment

(1) See Error flags (197)

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
32284-2^BKV^LN	EBV/BKV H (+) C	BKV^BKV^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	EBV/BKV L (+) C		Not used	
	Valid			
	Not used			
	(-) Ctrl			

📄 BKV control results

(1) See Error flags (197).

CMV - LIS mapping

CMV / CMV Dual is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	350

CMV sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
CMV	72493-0^CMV^LN

CMV tests

Target	Observation Identifier (OBX-3)
CMV	CMV^CMV^99ROC

CMV targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
CMV	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

CMV sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
72493-0^CMV^LN	CMV H (+) C	CMV^CMV^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	CMV L (+) C		Not used	
	Valid			
	Not used			

CMV control results
(1) See Error flags (197).

EBV - LIS mapping

EBV is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	200

EBV sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
EBV	43730-1^EBV^LN

EBV tests

Target	Observation Identifier (OBX-3)
EBV	EBV^EBV^99ROC

EBV targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
EBV	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

EBV sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
43730-1^EBV^LN	EBV/BKV H (+) C	EBV^EBV^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	EBV/BKV L (+) C		Not used	
	Valid			
	Not used			

 EBV control results
(1) See Error flags (197).

HBV - LIS mapping

HBV is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	200
		500
SERUM	SER^serum^HL70487	200
		500

HBV sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HBV	42595-9^HBV^LN

HBV tests

Target	Observation Identifier (OBX-3)
HBV	HBV^HBV^99ROC

HBV targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HBV	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

HBV sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
42595-9^HBV^LN	HxV H (+) C	HBV^HBV^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	HxV L (+) C		Not used	
	Valid Not used			

 HBV control results
(1) See Error flags (197).

HBV-RNA-RUO - LIS mapping

HBV-RNA-RUO is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUR^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	200
SERUM	SER^serum^HL70487	200

HBV-RNA-RUO sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HBV-RNA-RUO	HBVRNA^RUO^99ROC

HBV-RNA-RUO tests

Target	Observation Identifier (OBX-3)
HBV-RNA	HBVRNA^HBVRNA^99ROC

HBV-RNA-RUO targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HBV-RNA	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

HBV-RNA-RUO sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HBVRNA^HBVRNA^99ROC	HBV RNA H (+) C	HBVRNA^HBVRNA^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	HBV RNA L (+) C		Not used	
	(-) C		Valid Not used	

HBV-RNA-RUO control results
(1) See Error flags (197).

HCV - LIS mapping

HCV is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	200
		500
SERUM	SER^serum^HL70487	200
		500

☰ HCV sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HCV	11011-4^HCV^LN

☰ HCV tests

Target	Observation Identifier (OBX-3)
HCV	HCV^HCV^99ROC

☰ HCV targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HCV	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

☰ HCV sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
11011-4^HCV^LN	HxV H (+) C	HCV^HCV^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	HxV L (+) C		Not used	
	Valid			
	Not used			

HC HCV control results

(1) See Error flags (197).

HCV-PSC - LIS mapping

HCV-PSC is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
PSC	PSEPc^Plasma Separation Card^99ROC	850

☰ HCV-PSC sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HCV-PSC	11011-4^HCV-PSC^LN

☰ HCV-PSC tests

Target	Observation Identifier (OBX-3)
HCV	HCV^HCV^99ROC

☰ HCV-PSC targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HCV-PSC	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

☰ HCV-PSC sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
11011-4^HCV-PSC^LN	HxV H (+) C	HCV^HCV^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	HxV L (+) C		Not used	
	(-) C		Valid Not used	

HCVPSC control results

(1) See Error flags (197).

HDV - LIS mapping

HDV is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	200
SERUM	SER^serum^HL70487	200

HDV sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HDV	85512-2^HDV^LN

HDV tests

Target	Observation Identifier (OBX-3)
HDV	HDV^HDV^99ROC

HDV targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HDV	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

HDV sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
85512-2^HDV^LN	HDV H (+) C	HDV^HDV^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	HDV L (+) C		Not used	
	(-) C		Valid	
			Not used	

 HDV control results
(1) See Error flags (197).

HDV-RUO - LIS mapping

HDV-RUO is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	200
SERUM	SER^serum^HL70487	200

HDV-RUO sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HDV-RUO	HDV^RUO^99ROC

HDV-RUO tests

Target	Observation Identifier (OBX-3)
HDV	HDV^HDV^99ROC

HDV-RUO targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HDV	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

HDV-RUO sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HDV^RUO^99ROC	HDV H (+) C	HDV^HDV^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	HDV L (+) C		Not used	
	(-) C		Valid Not used	

HDV-RUO control results

(1) See Error flags (197).

HIV-1 - LIS mapping

HIV is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	200
		500

▣ HIV-1 sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HIV-1	70241-5^HIV^LN

▣ HIV-1 tests

Target	Observation Identifier (OBX-3)
HIV-1	HIV^HIV^99ROC

▣ HIV-1 targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HIV-1	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

▣ HIV-1 sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
70241-5^HIV^LN	HxV H (+) C	HIV^HIV^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	HxV L (+) C		Not used	
	(-) C		Valid	
			Not used	

📋 HIV-1 control results

(1) See Error flags (197).

HIV-1/2-Qual-DBS - LIS mapping

HIV1/2Qual-DBS is an example of a duplex qualitative assay with 2 channels (targets). For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
DRIED BLOOD SPOT	DBS^DriedBloodSpot^99ROC	850

▣ HIV-1/2-Qual-DBS sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HIV-1/2-Qual-DBS	85361-4^HIV-1-2-DBS^LN

▣ HIV-1/2-Qual-DBS tests

Target	Observation Identifier (OBX-3)
HIV-1	HIV 1^HIV 1^99ROC
HIV-2	HIV 2^HIV 2^99ROC

▣ HIV-1/2-Qual-DBS targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HIV-1	Reactive	RR
HIV-2	Non-Reactive	NR
	Not used	Error flag ⁽¹⁾

▣ HIV-1/2-Qual-DBS sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
85361-4^HIV-1-2-DBS^LN	HIV-1M/HIV-2 (+) C	HIV 1^HIV 1^99ROC HIV 2^HIV 2^99ROC	Valid Not used	VAL
	HIV-1O (+) C	HIV 1^HIV 1^99ROC		Error flag ⁽¹⁾
	(-) C	HIV 1^HIV 1^99ROC	Valid	
		HIV 2^HIV 2^99ROC	Not used	

■ HIV-1/2-Qual-DBS control results

(1) See Error flags (197).

HIV-1/2-Qual-Ser/Pla - LIS mapping

HIV1/2Qual-DBS is an example of a duplex qualitative assay with 2 channels (targets). For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PLASMA	PLAS^plasma^HL70487	500
SERUM	SER^serum^HL70487	500

▣ HIV-1/2-Qual-Ser/Pla sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HIV-1	85368-9^HIV-1-2-PlaSer^LN

▣ HIV-1-Ser/Pla-DBS tests

Target	Observation Identifier (OBX-3)
HIV-1	HIV 1^HIV 1^99ROC
HIV-2	HIV 2^HIV 2^99ROC

▣ HIV-1/2-Qual-Ser/Pla targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HIV-1	Reactive	RR
HIV-2	Non-Reactive	NR
	Not used	Error flag ⁽¹⁾

▣ HIV-1/2-Qual-Ser/Pla sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
85368-9^HIV-1-2-PlaSer^LN	HIV-1M/HIV-2 (+) C	HIV 1^HIV 1^99ROC HIV 2^HIV 2^99ROC	Valid Not used	VAL Error flag ⁽¹⁾
	HIV-1O (+) C	HIV 1^HIV 1^99ROC		
	(-) C	HIV 1^HIV 1^99ROC		
		HIV 2^HIV 2^99ROC		

■ HIV-1/2-Qual-Ser/Pla control results

(1) See Error flags (197).

HIV-PSC - LIS mapping

HIV-PSC is a quantitative single-channel assay. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
PSC	PSEPC^Plasma SeparationCard^99ROC	850

☰ HIV-PSC sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HIV-PSC	70241-5^HIVPSC^LN

☰ HIV-PSC tests

Target	Observation Identifier (OBX-3)
HIV-1	HIV^HIV^99ROC

☰ HIV-PSC targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HIV-PSC	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

☰ HIV-PSC sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
70241-5^HIVPSC^LN	HxV H (+) C	HIV^HIV^99ROC	Titer (value)	VAL Error flag ⁽¹⁾
	HxV L (+) C		Not used	
	(-) C		Valid Not used	

 HIV-PSC control results

(1) See Error flags (197).

RESP-4FLEX - LIS mapping

RESP-4FLEX is a qualitative multiplex assay.

 The RESP-4FLEX assay has no dedicated Universal Service Identifier, but each individual test has one. A customer specific group with an individual number of tests (1-4) in an arbitrary order can be requested in a single OML^O33 test-order by the LIS. Tests inside the initial group of the test-order must not be repeated later in the reflex test order.

For each test ORDER group, the software returns a result ORDER group in the OUL^R22 message. This is different compared to other multi channel assays (e.g. MPX) where the RESULT group is returned multiple times.

Sample type and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
Diluted in cobas [®] MIS	DCMIS^diluted in cobas MIS^99ROC	850

 RESP-4FLEX sample type and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
FluA	92142-9^FluA^LN
FluB	92141-1^FluB^LN
RSV	92131-2^RSV^LN
SCoV2	94309-2^SCoV2^LN

 RESP-4FLEX tests

Target	Observation Identifier (OBX-3)
FluA	FluA^FluA^99ROC
FluB	FluB^FluB^99ROC
RSV	RSV^RSV^99ROC
SCoV2	SCoV2^SCoV2^99ROC

 RESP-4FLEX targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Result interpretation (OBX-8-1)
All	Positive	POS
	Negative	NEG
	Not used	Error flag ⁽¹⁾

RESP-4FLEX sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
All	RESP-FLEX (+) C	FluA^FluA^99ROC	Valid Not used	VAL Error flag ⁽¹⁾
		FluB^FluB^99ROC		
	(-) Ctrl	RSV^RSV^99ROC		
		SCoV2^SCoV2^99ROC		

RESP-4FLEX control results

(1) See Error flags (197).

RESP-FLEX - LIS mapping

RESP-FLEX is a qualitative multiplex assay.

 The RESP-FLEX assay has no dedicated Universal Service Identifier, but each individual test has one. A customer specific group with an individual number of tests (1-12) in an arbitrary order can be requested in a single OML^O33 test-order by the LIS. Tests inside the initial group of the test-order must not be repeated later in the reflex test order.

For each test ORDER group, the software returns a result ORDER group in the OUL^R22 message. This is different compared to other multi channel assays (e.g. MPX) where the RESULT group is returned multiple times.

Sample type and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
Diluted in cobas [®] MIS	DCMIS^diluted in cobas MIS^99ROC	850

 RESP-FLEX sample type and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
FluA	85477-8^FluA^LN
FluB	85478-6^FluB^LN
RSV	85479-4^RSV^LN
SCoV2	94500-6^SCoV2^LN
AdV	91820-1^AdV^LN
MPV	91809-4^MPV^LN
EVRV	92807-7^EVRV^LN
CoV	88627-5^CoV^LN
hPIV1	91802-9^hPIV1^LN
hPIV2	91801-1^hPIV2^LN
hPIV3	91800-3^hPIV3^LN
hPIV4	91799-7^hPIV4^LN

 RESP-FLEX tests

Target	Observation Identifier (OBX-3)
FluA	FluA^FluA^99ROC
FluB	FluB^FluB^99ROC
RSV	RSV^RSV^99ROC
SCoV2	SCoV2^SCoV2^99ROC
AdV	AdV^AdV^99ROC
MPV	MPV^MPV^99ROC
EVRV	EVRV^EVRV^99ROC
CoV	CoV^CoV^99ROC
hPIV1	hPIV1^hPIV1^99ROC
hPIV2	hPIV2^hPIV2^99ROC
hPIV3	hPIV3^hPIV3^99ROC
hPIV4	hPIV4^hPIV4^99ROC

RESP-FLEX targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Result interpretation (OBX-8-1)
All	Positive	POS
	Negative	NEG
	Not used	Error flag ⁽¹⁾

RESP-FLEX sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
All	RESP-FLEX (+) C	FluA^FluA^99ROC	Valid Not used	VAL Error flag ⁽¹⁾
		FluB^FluB^99ROC		
		RSV^RSV^99ROC		
		SCoV2^SCoV2^99ROC		
		AdV^AdV^99ROC		
		MPV^MPV^99ROC		
		EVRV^EVRV^99ROC		
	(-) Ctrl	CoV^CoV^99ROC		
		hPIV1^hPIV1^99ROC		
		hPIV2^hPIV2^99ROC		
		hPIV3^hPIV3^99ROC		
		hPIV4^hPIV4^99ROC		

RESP-FLEX control results

(1) See Error flags (197).

SARS-CoV-2 Duo - LIS mapping

SARS-CoV-2 Duo is a qualitative and quantitative assay with 1 channel (target). For each test order, the software returns one OBX segment in an OUR^R22 message.

Sample type and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
VIRAL TRANSPORT MEDIA	VTM^Viral Transport Media^99ROC	400

☰ SARS-CoV-2 Duo sample type and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
SARS-CoV-2-Duo	97104-4^SARS-COV-2^LN

☰ SARS-CoV-2 Duo tests

Target	Observation Identifier (OBX-3)
SCoV2 Qual	SCoV2 Qual^SCoV2 Qual^99ROC
SCoV2 Quant	SCoV2 Quant^SCoV2 Quant^99ROC

☰ SARS-CoV-2 Duo targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Result interpretation (OBX-8-1)
SCoV2 Qual	Positive	POS
	Negative	NEG
	Not used	Error flag ⁽¹⁾
SCoV2 Quant	Titer value	VAL
	Not used	AT (Above Titer)
	Not used	BT (Below Titer)
	Target Not Detected	ND
	Not used	Error flag ⁽¹⁾

☰ SARS-CoV-2 Duo sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
97104-4^SARS-COV-2^LN	SARS-CoV-2 H (+) C		Titer (value)	
	SARS-CoV-2 L (+) C	SCoV2 Qual^SCoV2 Qual^99ROC	Not used	VAL
		SCoV2 Quant^SCoV2 Quant^99ROC	Valid	Error flag ⁽¹⁾
	(-) Ctrl		Not used	

 SARS-CoV-2 Duo control results

(1) See Error flags (197).

SCoV2-FluA/B - LIS mapping

SCoV2-FluA/B / SARS-CoV-2 & Influenza A/B v2 are qualitative assays with 4 channels (targets). For each test order, the software returns one OBX segment in an OUR^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (μl) (TCD-9-1)
VIRAL TRANSPORT MEDIA	VTM^Viral Transport Media^99ROC	400
COBAS PCR MEDIA SWAB	CPM^cobas PCR Media^99ROC	400

☰ SCoV2-FluA/B sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
SCoV2-FluA/B	95380-2^SCoVFlu^LN

☰ SCoV2-FluA/B tests

Target	Observation Identifier (OBX-3)
FluA	FluA^FluA^99ROC
SCoV2	SCoV2^SCoV2^99ROC
PanSarB	PanSarB^PanSarB^99ROC
FluB	FluB^FluB^99ROC

☰ SCoV2-FluA/B targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
FluA		
SCoV2	Positive	POS
	Negative	NEG
PanSarB	Not used	Error flag ⁽¹⁾
FluB		

☰ SCoV2-FluA/B sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
95380-2^SCoVFlu^LN	SCoV2-FluA/B (+) C	FluA^FluA^99ROC		
		SCoV2^SCoV2^99ROC	Valid	VAL
	(-) Ctrl	PanSarb^PanSarb^99ROC	Not used	Error flag ⁽¹⁾
		FluB^FluB^99ROC		

☐ SCoV2-FluA/B control results

(1) See Error flags (197).

SCoV2-QL - LIS mapping

SCoV2-QL is a qualitative assay with 2 channels (targets). For each test order, the software returns one OBX segment in an OUR^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
VIRAL TRANSPORT MEDIA	VTM^Viral Transport Media^99ROC	400
COBAS PCR MEDIA SWAB	CPM^cobas PCR Media^99ROC	400
SALIVA	SAL^Saliva^HL70487	850

SCoV2-QL sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
SCoV2-QL	94500-6^SCoV2QL^LN

SCoV2-QL tests

Target	Observation Identifier (OBX-3)
SCoV2	SCoV2^SCoV2^99ROC
PanSarB	PanSarB^PanSarB^99ROC

SCoV2-QL targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
SCoV2	Positive	POS
PanSarB	Negative	NEG
	Not used	Error flag ⁽¹⁾

SCoV2-QL sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
94500-6^SCoV2QL^LN	SCoV2-QL (+) C	SCoV2^SCoV2^99ROC	Valid	VAL
	(-) Ctrl	PanSarb^PanSarb^99ROC	Not used	Error flag ⁽¹⁾

SCoV2-QL control results

(1) See Error flags (197).

Sexual health / microbiology

This section lists the specific order, sample, and result codes used for each test.

In this section

BV/CV mapping (173)

CT/NG - LIS mapping (175)

HPV-GT - LIS mapping (177)

HPV-HR - LIS mapping (179)

MAI - LIS mapping (181)

MTB - LIS mapping (183)

MTB-RIF/INH - LIS mapping (185)

TV/MG - LIS mapping (187)

BV/CV mapping

BV/CV is a qualitative multi-channel assay with 2 targets. For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
SWAB	SWAB^Swab^99ROC	400

☰ BV/CV sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
BV/CV	92703-8^BV/CV^LN
BV	69568-4^BV^LN
CV	62460-1^CV^LN

☰ BV/CV tests

Target	Observation Identifier (OBX-3)
BV	BV^BV^99ROC
CV	CV^CV^99ROC

☰ BV/CV targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
BV, CV	Positive	POS
	Negative	NEG
	Not used	Error flag ⁽¹⁾

☰ BV/CV sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
92703-8^BV/CV^LN	BV/CV (+) C	BV^BV^99ROC	Valid	VAL
	(-) Ctrl	CV^CV^99ROC	Not used	Error flag ⁽¹⁾

 BV/CV control results
(1) See Error flags (197).

CT/NG - LIS mapping

CT/NG is a qualitative assay with 2 channels (targets). For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
SWAB	SWAB^Swab^99ROC	400
URINE	UR^Urine^HL70487	850
PRESERVCYT	PCYT^preservCyt^99ROC	400

CT/NG sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
CT/NG	72828-7^CT/NG^LN
CT	21613-5^CT^LN
NG	24111-7^NG^LN

CT/NG tests

Target	Observation Identifier (OBX-3)
CT	CT^CT^99ROC
NG	NG^NG^99ROC

CT/NG targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
CT	Positive	POS
NG	Negative	NEG
	Not used	Error flag ⁽¹⁾

CT/NG sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
72828-7^CT/NG^LN	CT/NG (+) C	CT^CT^99ROC	Valid	VAL
	(-) Ctrl	NG^NG^99ROC	Not used	Error flag ⁽¹⁾

CT/NG control results

(1) See Error flags (197).

HPV-GT - LIS mapping

HPV-GT is a qualitative assay with three channels (targets). For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
Cell Collection Medium PLUS	CCM^Cell Collection Medium PLUS^99ROC	400
PRESERV CYT	PCYT^preservCyt^99ROC	400
RCCM	RCCM^RocheCellCollectionMedia^99ROC	400
Self, vaginal - RCCM/PC	SVAG^self, vaginal - RCCM/PC^99ROC	400
Self, vaginal (for US only)	SVAL^self, vaginal^99ROC	400
SUREPATH	SPATH^SurePath^99ROC	400

HPV-GT sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HPV-GT	71432-9^HPV-GT^LN

HPV-GT tests

Target	Observation Identifier (OBX-3)
Other HR HPV	Other HR HPV^Other HR HPV^99ROC
HPV 16	HPV 16^HPV 16^99ROC
HPV 18	HPV 18^HPV 18^99ROC

HPV-GT targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
Other HR HPV	Positive	POS
HPV 16	Negative	NEG
HPV 18	Not used	Error flag ⁽¹⁾

HPV-GT sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
71432-9^HPV-GT^LN	HPV (+) C	Other HR HPV^Other HR HPV^99ROC	Valid	VAL
	(-) Ctrl	HPV 16^HPV 16^99ROC	Not used	Error flag ⁽¹⁾
		HPV 18^HPV 18^99ROC		

HPV-GT control results
(1) See Error flags (197).

HPV-HR - LIS mapping

HPV-HR is a qualitative assay with one channel (target). For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
Cell Collection Medium PLUS	CCM^Cell Collection Medium PLUS^99ROC	400
PRESERVCYT	PCYT^preservCyt^99ROC	400
RCCM	RCCM^RocheCellCollectionMedia^99ROC	400
Self, vaginal - RCCM/PC	SVAG^self, vaginal - RCCM/PC^99ROC	400
Self, vaginal (for US only)	SVAG^self, vaginal^99ROC	400
SUREPATH	SPATH^SurePath^99ROC	400

HPV-HR sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
HPV-HR	82675-0^HPV-HR^LN

HPV-HR tests

Target	Observation Identifier (OBX-3)
HPV-HR	HR HPV^HR HPV^99ROC

HPV-HR targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
HPV-HR	Positive	POS
	Negative	NEG
	Not used	Error flag ⁽¹⁾

HPV-HR sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
82675-0^HPV-HR^LN	HPV (+) C	HR HPV^HR HPV^99ROC	Valid	VAL
	(-) Ctrl		Not used	Error flag ⁽¹⁾

HPV-HR control results

(1) See Error flags (197).

MAI - LIS mapping

MAI is a qualitative assay with 2 channels (targets). For each test order, the software returns one OBX segment in an OUR^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
RAW SPUTUM	SPTR^rawSputum^99ROC	850
SEDIMENT	SED^Sediment^99ROC	850

MAI sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
MAI	85581-7^MAI^LN

MAI tests

Target	Observation Identifier (OBX-3)
MIN	MIN^MIN^99ROC
MAV	MAV^MAV^99ROC

MAI targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
MIN	Positive	POS
MAV	Negative	NEG
	Not used	Error flag ⁽¹⁾

MAI sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
85581-7^MAI^LN	MAI (+) C	MIN^MIN^99ROC	Valid	VAL
	(-) Ctrl	MAV^MAV^99ROC	Not used	Error flag ⁽¹⁾

MAI control results

(1) See Error flags (197).

MTB - LIS mapping

MTB is a qualitative assay with 1 channel (target). For each test order, the software returns one OBX segment in an OUR^R22 message..

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
RAW SPUTUM	SPTR^rawSputum^99ROC	850
SEDIMENT	SED^Sediment^99ROC	850

MTB sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
MTB	85362-2^MTB^LN

MTB tests

Target	Observation Identifier (OBX-3)
MTB	MTB^MTB^99ROC

MTB targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
MTB	Positive	POS
	Negative	NEG
	Not used	Error flag ⁽¹⁾

MTB sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
85362-2^MTB^LN	MTB (+) C	MTB^MTB^99ROC	Valid	VAL
	(-) Ctrl		Not used	Error flag ⁽¹⁾

MTB control results

(1) See Error flags (197).

MTB-RIF/INH - LIS mapping

MTB-RIF/INH is a qualitative assay with 4 channels (targets). For each test order, the software returns one OBX segment in an OUR^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
RAW SPUTUM	SPTR^rawSputum^99ROC	850
SEDIMENT	SED^Sediment^99ROC	850

MTB-RIF/INH sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
MTB-RIF/INH	89876-7^RIFINH^LN

MTB-RIF/INH tests

Target	Observation Identifier (OBX-3)
INH	INH^INH^99ROC
RIF1	RIF1^RIF1^99ROC
RIF2	RIF2^RIF2^99ROC
RIF3	RIF3^RIF3^99ROC

MTB-RIF/INH targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
INH		
RIF1	Positive	POS
RIF2	Negative	NEG
RIF3	Not used	Error flag ⁽¹⁾

MTB-RIF/INH sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
89876-7^RIFINH^LN	MTB-RIF/INH (+) C	INH^INH^99ROC	Valid	VAL
		RIF1^RIF1^99ROC	Not used	Error flag ⁽¹⁾
	(-) Ctrl	RIF2^RIF2^99ROC		
		RIF3^RIF3^99ROC		

 MTB-RIF/INH control results
(1) See Error flags (197).

TV/MG - LIS mapping

TV/MG is a qualitative assay with 2 channels (targets). For each test order, the software returns one OBX segment in an OUL^R22 message.

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
PRESERVCYT	PCYT^preservCyt^99ROC	400
SWAB	SWAB^Swab^99ROC	400
URINE	UR^Urine^HL70487	850
MEATAL SWAB	MEATAL^Meatal^99ROC	850

TV/MG sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
TV/MG	86949-5^TV/MG^LN
TV	69937-1^TV^LN
MG	69935-5^MG^LN

TV/MG tests

Target	Observation Identifier (OBX-3)
TV	TV^TV^99ROC
MG	MG^MG^99ROC

TV/MG targets

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
TV	Positive	POS
	Negative	NEG
MG	Not used	Error flag ⁽¹⁾

TV/MG sample result codes for OBX segment

(1) See Error flags (197).

Control results

These values are sent in a control message.

Universal Service Identifier (OBR-4)	Control Name (INV-1-1)	Observation Identifier (OBX-3)	Observation value (OBX-5)	Interpretation code (OBX-8-1)
86949-5^TV/MG^LN	TV/MG (+) C	TV^TV^99ROC	Valid	VAL
	(-) Ctrl	MG^MG^99ROC	Not used	Error flag ⁽¹⁾

TV/MG control results

(1) See Error flags (197).

UCAP

This section lists the specific order, sample, and result codes used for each test.

In this section

UCAP (190)

UCAP

Sample types and input volume

Name	Sample Type (SPM-4)	Volume (µl) (TCD-9-1)
COBAS PCR MEDIA SWAB	CPM^cobas PCR Media^99ROC	400
Diluted in cobas MIS	DCMIS^diluted in cobas MIS^99ROC	850
PLASMA	PLAS^plasma^HL70487	200, 350, 500, 800
PRESERVCYT	PCYT^preservCyt^99ROC	400
ROCHE CELL COLLECTION MEDIA	RCCM^RocheCellCollectionMedia^99ROC	400
Raw sputum	SPTR^rawSputum^99ROC	850
Sediment	SED^Sediment^99ROC	850
SERUM	SER^serum^HL70487	200, 500, 850
SWAB	SWAB^Swab^99ROC	400
U_Alcohol-based sample	UCAlcS^UC_Alcohol-based sample^99ROC	150, 200, 400, 850
U_Buffer-based sample	UCBufS^UC_Buffer-based sample^99ROC	150, 200, 400, 850
U_Sample with swab	UCSwabS^UC_Sample with swab^99ROC	150, 200, 400, 850
U_Simple sample	UCSimpS^UC_Simple sample^99ROC	150, 200, 350, 500, 850
URINE	UR^Urine^HL70487	400, 850
WHOLE BLOOD	BLD^Whole Blood^HL70487	500
VIRAL TRANSPORT MEDIA	VTM^Viral Transport Media^99ROC	400

☰ UCAP sample types and input volume

Tests and targets

Test	Universal Service Identifier (OBR-4 / TCD-1)
xyz ⁽¹⁾	7 digit number followed by ^UCAP^99ROC (e.g. 0000041^UCAP^99ROC)

☰ UCAP tests

(1) xyz is defined in the Cobas Development Channel Tool (CDC)

Target	Observation Identifier (OBX-3)
Target 1 ⁽¹⁾	Target 1^Target 1^99ROC
Target 2 ⁽²⁾	Target 2^Target 2^99ROC
Target 3 ⁽²⁾	Target 3^Target 3^99ROC
Target 4 ⁽²⁾	Target 4^Target 4^99ROC

☰ UCAP targets

(1) Name defined in the Cobas Development Channel Tool (CDC). "Target 1" is just an example.

(2) Up to 4 targets can be defined

Sample result codes for OBX segment

Target	Observation value (OBX-5)	Interpretation code (OBX-8-1)
Target 1 ⁽¹⁾		
Target 2 ⁽³⁾	Reactive	RR
	Non-Reactive	NR
Target 3 ⁽³⁾	Not used	Error flag ⁽²⁾
Target 4 ⁽³⁾		

UCAP sample result codes for OBX segment

(1) Name defined in the Cobas Development Channel Tool (CDC). "Target 1" is just an example.

(2) See Error flags (197).

(3) Up to 4 targets can be defined

Examples

In this chapter

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Examples. 195

Examples

For example communication messages, please contact your Roche Service representative.

Error flags

In this chapter

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Error flags 199

Error flags



Error flags are dynamic and may change depending on the installed ASAP.

LIS flag	Description
C01	Calculation error
C01[H1...H5]	Calculation error for target 1 to 5
C02	Data cannot be analyzed.
C02[H1...H5]	Data cannot be analyzed for target 1 to 5
P02T	Clot detected while aspirating/mixing sample in sample tube. Sample not transferred.
P03T	Clot detected while dispensing sample.
P04T	Insufficient control volume detected during sample transfer.
P05T	Droplet detected during sample transfer.
P06P	Error while dispensing reagent due to insufficient volume.
P06T	Liquid level detection failed during sample transfer.
P07P	Insufficient supernatant removed.
P07T	Clot detected while aspirating/dispensing a control.
P08T	Pipetting error occurred while aspirating control. Control not transferred.
P09P	Processing temperature in heating station out of range.
P09T	Risk of cross-contamination.
P10P	Maximum allowed time for sample transfer exceeded.
P10T	Maximum allowed time for sample transfer exceeded.
P11A	Temperature profile error.
P12A	Fluorescence signal saturated
P13A	Camera temperature out of range.
P13P	Camera temperature out of range.
P14A	Light source intensity out of range.
P14P	Consumable dropped.
P15A	Heated lid temperature out of range.
P15P	Consumable pickup/release error
P16A	Maximum time defined in the PCR profile exceeded.
P16P	Pipette tip could not be picked up by processing transfer head during aspiration.
P17A	Camera connection error
P17P	Sealing station could not be activated.
P18P	Thermal cycler could not be activated.
P19A	Error of the photometer.
P19P	Incubation temperature of heating station cannot be set.
P20P	Sealing station error.
P21P	Maximum allowed time for handling filled and unsealed amplification plate exceeded.
P22P	Maximum allowed time for handling filled and sealed amplification plate exceeded.
P23P	Sealing station loading error.
P24P	Processing transfer head: Bottom detection failed.
Q03D	Result validation error
Q04D	Result enrichment error

Error flags

Roche Diagnostics

x800 Data Manager · Software version 2.1 and higher · Host Interface Manual · Version 5.3

LIS flag	Description
Q05D	Result validation failed because of an invalid positive control
Q06D	Result validation failed because of an invalid negative control
Q07D	Reagent cassette not validated.
U04	User aborted test order
U06T	Pipetting error while aspirating sample.
U07T	Control mini rack RFID parameter error
U08	Unexpected opening of reagent aspiration arm was detected.
U12M	Sample ID entered manually by keyboard or scanned using hand-held barcode reader.
Y02T	Clot detected while aspirating/dispensing a control.
Y03	Hardware error during carrier transfer to a transfer position.
Y04P	Hardware error in processing module.
Y06A	Hardware error in analytic module.
Y07	Required hardware module not available to continue the process.
Y11	A hardware error occurred. Result cannot be guaranteed.
Y13	Mechanical error of MGP cassette shaker.
Y14	Mechanical error of sample transfer head.
Y15	Mechanical error of reagent transfer head.
Y16	Mechanical error of processing transfer head.
Y17	Mechanical error of the separation station.
Y18	Mechanical error of heating station.
Y19	Mechanical error of downholders in the processing unit
Y20M	Instrument software error.
Y21A	Mechanical error of the thermal cyclers.
Y31D	Work order was interrupted by a power failure.
Y32D	Processing order aborted manually.
Y44	Temperature in the reagent supply unit out of range.
Y45	Maximum allowed temperature for control mini rack drawer exceeded.
Y46	Temperature in the interim reagent storage exceeded the upper limit.
Y47	Onboard temperature dropped below lower limit.
Y48	Required hardware module not available to continue the process.
Y49P	Temperature of the amplification plate park position out of range.

 Error flags

Communication log

In this chapter

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Communication log. 203

Communication log

The Data Manager stores all communications into a log file (`host-handler.txt`) that is part of the standard problem report.

WARNING

Disclosure of confidential data in log file

Unauthorized users can view confidential data stored in the log file.

- ▶ Ensure that the log file is protected from any unauthorized access.
 - ▶ Restrict physical access to the log file.
 - ▶ Read the respective safety information on data security provided along with this product.
-

Glossary

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Glossary

ASAP Array of files that contain the parameters and specifications of a specific test. It includes definitions for the robotics, calculation packages for the result calculation and libraries for the plausibility calculation.

GUID A 128-bit hexadecimal number created by the software to identify a message in the transmission record (e.g. 3F2504E0-4F89-11D3-9A0C-0305E82C3301).

LIS (Laboratory Information System) Host computer connected to the x800 Data Manager. It holds the patient Information and communicates with the x800 Data Manager using the HL7 protocol.

LOINC Preferred code set of HL7 for laboratory test names. LOINC validated by HL7 is effective globally. Development LOINC is system-specific and not validated by HL7 (e.g. created by Roche for x800 Data Manager).

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